A Fast Algorithm for Subspace Convolution

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Abstract

A cornerstone result in Boolean lattice theory is the subset-zeta transform, which computes the f-weighted subset summation problem $(Zf)(B) = \sum_{A \subseteq B} f(A)$ for all $B \subseteq [n]$ in $O(N \log N)$ additions instead of the naïve $O(3^n) = O(N^{\log_2 3})$, where $N = 2^n$.

This paper studies the algebraic analogue of this problem–the *subspace summation* problem–where subsets are replaced by linear *subspaces* in $\mathbb{F}_q^n$. Surprisingly, we show that this problem admits a similar $\Theta_q(N_q \log N_q)$-time algorithm, where N_q is the number of subspaces. This improves substantially (in both theory and practice) over lattice-based algorithms ([BHK+16, KKW16]), which are known to be optimal for addition-only algorithms over $\mathbb{R}$. We prove similar runtimes for inversion, rank-constrained sums and span convolution. We describe two main applications:

- **Fast linear algebra on Grassmann graphs:** Matrix–vector multiplication for arbitrary distance-invariant kernels on N-vertex Grassmann graphs in $O_q(N \log N)$ arithmetic operations, together with fast spectral projections and exact solutions of the corresponding linear systems.
- **Graph coloring of low binary cut-rank graphs:** An $n^{2^{r^2/4+O(1)}}$-time algorithm for computing the chromatic number and chromatic polynomial of an n-vertex graph, given a rank-decomposition of width r. We also give quantitative reductions from satisfiability and monotone constraint problems.

Experiments show speedups in measured running time of up to 18 over the covering-pair algorithm for the zeta transform, 190 over Poisson summation for rank-constrained sums, and 22 over the distance-matrix recurrence for Grassmann kernel multiplication. Lean 4 formalizations of the transform identities, arithmetic operation bounds, and graph-coloring constructions accompany the paper [PW26].

*Research supported by ERC Starting grant CODY 101039914, and ISF grant 3011005535.
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1 Introduction

A fundamental problem in Boolean lattice theory is *subset summation*: given a coefficient $f(A) \in \mathbb{R}$ for every $A \subseteq [n]$, compute $(\mathsf{Z}f)(B) = \sum_{A \subseteq B} f(A)$ for every $B \subseteq [n]$, where $f, \mathsf{Z}f \in \mathbb{R}^N$, and $N = 2^n$. For a given subset B, the sum has $2^{|B|}$ terms and takes at most $N-1$ additions. Computing all sums separately enumerates $\sum_B 2^{|B|} = 3^n = N^{\log_2 3}$ terms. The classical *fast zeta transform* (see [KKW16, Section 1.1]) uses only $O(n2^n) = O(N \log N)$ *additions* to evaluate all sums. A similar algorithm applies to the inverse subset-sum problem, via the Möbius transform [KKW16], which is the inverse of the zeta transform. These transforms led to profound algorithmic speedups for a variety of combinatorial problems, including the Chromatic Number, Hamiltonian path and TSP problems [BHK09, BHKK07, Bjo10], Steiner tree decompositions [BHKK07], perfect matchings and cycle covers [BHK09].

We study the analogous problem for *linear subspaces*. Throughout, $\mathbb{F}_q$ is a finite field of fixed order $q = p^m$, where p is prime, and $\mathrm{Gr}_q(n)$ is the set of all subspaces of $\mathbb{F}_q^n$. Given $f : \mathrm{Gr}_q(n) \to \mathcal{R}$, where $\mathcal{R}$ is a commutative coefficient ring, compute

$$(\mathsf{Z}_n f)(W) = \sum_{U \leq W} f(U), \qquad W \in \mathrm{Gr}_q(n). \tag{1}$$

Here $U \leq W$ means that U is a linear subspace of W. Thus f and $\mathsf{Z}_n f$ each have $N_q(n) = |\mathrm{Gr}_q(n)|$ entries. The matrix Z_n has entry $\mathsf{Z}_n(W, U) = 1$ if $U \leq W$ and zero otherwise. For example, over $\mathbb{Q}$, taking f to be the indicator of a family of subspaces makes $(\mathsf{Z}_n f)(W)$ count the members of that family contained in W. The inverse of Z_n is called the Möbius transform, and the transpose $\mathsf{Z}_n^{\mathsf{T}}$ sums over superspaces.

The subset algorithm passes information from each set to its immediate supersets, of which there are at most n. A subspace $U \leq \mathbb{F}_q^n$ has $(q^{n-\dim U} - 1)/(q-1)$ immediate superspaces, making the same approach much more expensive – the number of subspaces in $\mathbb{F}_q^n$ is $N_q(n) = \Theta_q(q^{\lfloor n^2/4 \rfloor})$ (see Appendix A), whereas the number of *covering pairs* $U < W$ with $\dim W = \dim U + 1$ is $\Theta_q(N_q(n) q^{n/2})$ (see (27)). The lattice algorithms of [KKW16, BHK+16, Theorem 1.3] therefore take $\Theta_q(N_q(n) q^{n/2}) = \Theta_q\left(N_q(n)^{1+1/\sqrt{\log_q N_q(n)}}\right)$ arithmetic operations, proportional to the number of these pairs. We call this approach the *covering-pair algorithm*. Over $\mathbb{R}$, Kennes [Ken92, Theorem 6] proves a matching lower bound for algorithms using only additions.

Our algorithm avoids this addition-only lower bound by using subtraction and scalar multiplication. The resulting cancellations let us sum over contained subspaces without listing all these pairs. The core idea behind our *fast subspace transform* is a recursive hyperplane cutting technique, following Srinivasan's character decomposition [Sri14]. This reduces affine containment of *crossing subspaces* to a translation-invariant sum, allowing an ordinary FFT to decompose the subspace sum problem into lower-dimensional subspace summation problems in the frequency domain.

1.1 Our Results

Let $\mathrm{Gr}_q(n, j)$ be the set of all j-dimensional subspaces in $\mathbb{F}_q^n$. The number of subspaces of dimension at most k is

$$M_q(n, k) = \sum_{j=0}^{\min(k,n)} |\mathrm{Gr}_q(n, j)|, \qquad M_q(n, n) = N_q(n).$$

The truncated containment matrix $\mathsf{Z}_{n,k}$ is the $M_q(n,k) \times M_q(n,k)$ matrix with entries $\mathbf{1}[U \leq W]$, where $\mathbf{1}[\cdot]$ is one when its condition holds and zero otherwise, and both indices have dimension at

most k. Ordering by dimension makes it triangular with diagonal one. For $k = n$, this is the full transform Z_n. Its transpose sums over superspaces in the same table. Our algorithms require the following assumption on the coefficient ring $\mathcal{R}$.

Theorem 1 (Fast Subspace Summation)**.** *Over any commutative ring in which p is invertible, each of $\mathsf{Z}_{n,k}$, $\mathsf{Z}_{n,k}^{-1}$, $\mathsf{Z}_{n,k}^{\mathsf{T}}$, and $(\mathsf{Z}_{n,k}^{\mathsf{T}})^{-1}$ can be applied in $O_q(nkM_q(n,k))$ arithmetic operations for $k \le n$. In particular, the cost is $O_q(M_q(n,k)\log M_q(n,k))$.*

Corollary: Fast Subspace Convolution Zeta transforms also turn convolution into pointwise multiplication. For two subspace coefficient functions f, g on $\mathrm{Gr}_q(n)$, define their *span convolution* by

$$(f \star g)(W) = \sum_{U+V=W} f(U)g(V),$$

where the sum is over ordered pairs of subspaces. Since $U + V \le W$ exactly when $U, V \le W$, $\mathsf{Z}_n(f \star g) = (\mathsf{Z}_n f)(\mathsf{Z}_n g)$, with pointwise multiplication on the right. By theorem 1, two transforms and one inverse compute $f \star g$ in $O_q(N_q(n)\log N_q(n))$ arithmetic operations.

Low-Rank Summation One may instead have weights only on a-dimensional subspaces and want sums only on b-dimensional subspaces. For example, given weights on one-dimensional subspaces (projective points), compute their sums in every hyperplane. In this example, the input and output tables have $q^{O(n)}$ entries, whereas the full subspace table has $q^{\Theta(n^2)}$ entries. Consequently, applying the full transform takes $q^{\Theta(n^2)}$ work, whereas theorem 2 below brings this down to $q^{O(n)}$. For a coefficient ring $\mathcal{R}$ and $0 \le a \le b \le n$, define the *incidence transform* by

$$(\mathsf{I}_n^{a,b} f)(W) = \sum_{\substack{U \le W \\ \dim U = a}} f(U), \qquad f : \mathrm{Gr}_q(n,a) \to \mathcal{R}, \quad W \in \mathrm{Gr}_q(n,b).$$

Its matrix has $|\mathrm{Gr}_q(n,b)|$ rows and $|\mathrm{Gr}_q(n,a)|$ columns. The transpose sends weights on b-dimensional subspaces to sums over those containing each a-dimensional subspace.

Theorem 2. *Under the coefficient assumption of theorem 1, $\mathsf{I}_n^{a,b}$ and its transpose take $O_q(S\log S)$ arithmetic operations, where $S = |\mathrm{Gr}_q(n,a)| + |\mathrm{Gr}_q(n,b)|$ is the total input and output length.*

For $(n,a,b) = (4t,t,3t)$, this bound is optimal in the bounded-coefficient model over $\mathbb{C}$ (see Appendix F). Constructing the subspace bases and coordinate changes used to index these tables takes table size times $\mathrm{poly}(n)$ additional work. Appendix D gives the construction and exact bounds.

1.1.1 Applications

Fast linear algebra on Grassmann graphs A *geometric kernel* assigns a weight $K(x_i, x_j)$ to each pair of points $x_i \in \mathbb{R}^\ell$. A standard example is the radial kernel $K(x,y) = f(\|x-y\|_2^2)$. For N points $x_i \in \mathbb{R}^\ell$, K implicitly defines an $N \times N$ matrix using only $N\ell$ coordinates and the function f. If each coordinate uses $\mathrm{poly}(\log(N\ell))$ bits and f has a bounded-size description, this is $\widetilde{O}(N\ell)$ information bits. As such, one could hope to *multiply* K by a vector (or perform other linear-algebraic operations) in subquadratic time ($\ll N^2$). The celebrated Fast Multipole Method of Greengard and Rokhlin [GR87] gives such approximate matrix–vector products for suitable kernels. For Gaussian kernels, Greengard and Strain developed the Fast Gauss Transform [GS91]. Alman, Chu, Schild, and Song [ACSS20] initiate a systematic study of fast linear algebra on geometric

graphs, including approximate matrix–vector multiplication, spectral sparsification, and Laplacian solving.

We study a finite-geometric analogue: kernels determined by intersection dimensions on the full Grassmannian. Let $V = \mathrm{Gr}_q(n,k)$, $N = |V|$, and $d = \min(k, n-k)$. The Grassmann graph $J_q(n,k)$ has vertex set V, where two vertices U, W are adjacent exactly when $\dim(U \cap W) = k-1$. Their graph distance is $\mathrm{dist}(U,W) = k - \dim(U \cap W)$, and its maximum is d. An *intersection kernel* is an $N \times N$ matrix specified by coefficients $a_0, \ldots, a_d \in \mathcal{R}$.[1] Given weights $f : V \to \mathcal{R}$, we want to compute

$$K(U,W) = a_{\mathrm{dist}(U,W)}, \qquad (Kf)(U) = \sum_{W \in V} a_{\mathrm{dist}(U,W)} f(W).$$

Our fast subspace transform computes this product without constructing all N^2 entries:

Theorem 3 (Fast multiplication on the Grassmann graph)**.** *Under the coefficient assumption of theorem 1, given $a_0, \ldots, a_d \in \mathcal{R}$ and $f : V \to \mathcal{R}$, the product Kf can be computed in $O_q(N \log N)$ arithmetic operations in $\mathcal{R}$.*

Earlier covering-pair transforms [KKW16] give $O_q(q^d N)$ operations. Our bound replaces q^d by $\log N$, a substantial saving when q^d is much larger.

For example, choosing $a_j = \mathbf{1}[j = i]$ gives the distance matrix D_i, which sums weights at distance i: $(D_i f)(U) = \sum_{W \in V : \mathrm{dist}(U,W) = i} f(W)$. In particular, D_1 is the adjacency matrix and sums neighbors' weights. If f is the indicator of a nonempty family $\mathcal{C} \subseteq V$, computing over $\mathbb{Q}$ counts its members at each distance from U. The smallest i with $(D_i f)(U) > 0$ is the distance to its nearest member.

We also compute all projections of f onto eigenspaces of D_1 over $\mathbb{Q}$ or $\mathbb{C}$ in $O_q((d+1)N \log N)$ operations. Over any field of characteristic different from p, the same bound tests whether K is nonsingular and, if so, solves $Kx = b$. For the spectral task, Maslen, Orrison, and Rockmore [MOR03, Theorem 5.8] use $O(d[d]_q N)$ operations, where $[d]_q = (q^d - 1)/(q-1)$. Our bound improves the dependence on d when $[d]_q$ is much larger than $\log N$, though their bound can be smaller for small d. Section 5 proves the multiplication bound, and Appendix C gives the distance sums and further applications.

Experiments. The constants hidden in our asymptotic bounds could make our algorithms slower in practice, so we tested them against the comparison algorithms in Table 1. Our measured running times show speedups of up to 18 for the zeta transform Z_n, 190 for incidence $\mathsf{I}_n^{a,b}$, and 22 for Grassmann kernel multiplication. On these instances, the operation counts for the comparison algorithms reach 4.1, 170, and 4.6 times ours, respectively. Both the zeta and incidence algorithms also handle instances for which the comparison methods exceed the available memory. These results are discussed in more detail in Section 7.

Graph coloring of low binary cut-rank graphs Subspace containment also describes which color classes can share a color across a graph cut. Our transform sums over colorings that can be combined across the cut, and the same subspace representation leads to an improved algorithm for coloring the whole graph. For a graph G with adjacency matrix A_G, the *binary cut-rank* of a cut is

$$\rho_G(X) := \mathrm{rank}_{\mathbb{F}_2} A_G[X, V(G) \setminus X].$$

[1] Embed U as $x_U = (\mathbf{1}[L \le U])_{L \in \mathrm{Gr}_q(n,1)}$. Then $\|x_U - x_W\|_2^2 = 2(q^k - q^{\dim(U \cap W)})/(q-1)$, so intersection kernels are radial kernels on these vectors.

Across a cut of binary cut-rank r, vertices can be labeled by columns $p_x, q_y \in \mathbb{F}_2^r$ so that xy is an edge exactly when $p_x^{\mathsf{T}} q_y = 1$. A color class on either side is represented by the *span* of its labels, called its *profile*. The number of profiles is at most the number of subspaces of $\mathbb{F}_2^r$, which is

$$N_2(r) := 2^{r^2/4+O(1)}$$

(see Section 6). Two color classes on opposite sides can share a color precisely when their spans U, V are orthogonal, or equivalently $V \le U^\perp$. Given weights $g(V)$ for all $V \le \mathbb{F}_2^r$, theorem 1 computes $\sum_{V \le U^\perp} g(V) = (\mathsf{Z}_r g)(U^\perp)$ for every U. Moreover, proposition 1 computes the corresponding sums for several colors in time nearly linear in the number of input weights.

To color the whole graph, we need to account for the number of classes having each of the $N_2(r) = 2^{r^2/4+O(1)}$ possible subspace profiles. Our algorithm combines these counts without applying the transform. A *rank-decomposition* is a subcubic tree whose leaves are in bijection with $V(G)$, and its width is the maximum binary cut-rank over the cuts induced by deleting a tree edge. The *rank-width* $\mathrm{rw}(G)$ is the minimum width over all rank-decompositions. A *linear layout* is a vertex ordering $\pi = (v_1, \ldots, v_n)$, whose width is $r(\pi) = \max_{1 \le i < n} \rho_G(\{v_1, \ldots, v_i\})$. The *linear rank-width* $\mathrm{lrw}(G)$ is the minimum over these orderings. We assume that the decomposition or layout is supplied.

For a fixed number c of colors, Bui-Xuan, Telle, and Vatshelle [BTV10] give a $2^{O(cr^2)} n^{O(1)}$-time algorithm. For unrestricted palettes, Ganian, Hliněný, and Obdržálek [GHO13, Theorems 4.1 and 4.7] compute the chromatic number and chromatic polynomial in $n^{2^{r^2/2+O(r)}}$ time by recording the multiplicities of subspace profiles and the ways to pair color classes. We improve the exponent by processing these pairings one profile at a time:

Theorem 4 (informal)**.** *Given an n-vertex graph G and a rank-decomposition of width r, the chromatic number, an optimal coloring, and the chromatic polynomial of G can be computed in time*

$$n^{O(N_2(r))} = n^{2^{r^2/4+O(1)}}.$$

Assuming ETH [IP01], the leading coefficient $1/4$ is optimal, even for supplied linear layouts.

The algorithm takes $n^{2^{r^2/4+O(1)}}$ time, whereas ETH rules out even $n^{2^{r^2/4-17r}}$ time. See theorems 6 and 8 for the precise bounds.[2]

For every fixed number $c \ge 3$ of colors, ETH also excludes $2^{o(r^2)} n^{O(1)}$ time, even with a supplied linear layout (theorem 9). Under this hypothesis, the bound matches the exponent order of the algorithm of [BTV10] and answers the coloring question in [BKN23, Section 8, full version]. Appendix H gives the corresponding fixed-color counting lower bound under counting ETH.

Formal verification. The companion repository [PW26] contains Lean 4 proofs of the transform identities, arithmetic operation bounds, and graph-coloring constructions, using Mathlib, together with an axiom audit. The coloring lower bounds are formalized as conditional running-time implications. Index construction, some bit-cost arguments, and the machine-model running times of the reductions remain outside the formalization. For the fixed-palette reduction, the formalization uses one clause-local group per literal occurrence. The repository documents this variant, the cost-model assumptions, and the remaining unformalized claims.

[2] A recent preprint reports subexponential algorithms for deciding and counting 3-SAT [Cla26]. If correct, these refute ETH and counting ETH. Our reductions remain unconditional implications between running times, as explained in section 1.2.

1.2 Technical overview

This section provides a streamlined overview of the proof of Theorem 1, its extensions, and the applications in Section 1.1.

1.2.1 Fast zeta transform

A natural approach to achieving a fast zeta transform is to partition $\mathrm{Gr}_q(n)$ into two sets and apply a smaller zeta transform to each. To make this precise, call a set $\mathcal{S}$ of subspaces an *induced copy* of $\mathrm{Gr}_q(i)$ if there is a bijection $\phi : \mathcal{S} \to \mathrm{Gr}_q(i)$ such that

$$U_1 \leq U_2 \quad \iff \quad \phi(U_1) \leq \phi(U_2), \qquad U_1, U_2 \in \mathcal{S}.$$

If the two sets were induced copies of $\mathrm{Gr}_q(i)$ and $\mathrm{Gr}_q(j)$, with $i, j < n$, and had no containments between them, ordering the input and output coordinates by these sets would give

$$\mathsf{Z}_n f = \mathrm{diag}(\mathsf{Z}_i, \mathsf{Z}_j) f.$$

We could then apply the smaller transforms independently and repeat this decomposition recursively. However, no partition into two nonempty sets has this property.[3]

Instead, we seek a partition of $\mathrm{Gr}_q(n)$ in which one set is an induced copy of a smaller subspace lattice on which one smaller transform is immediate, and the other set requires extra care. Fix the hyperplane $H = \mathbb{F}_q^{n-1} \times \{0\}$. The subspaces contained in H form an induced copy of $\mathrm{Gr}_q(n-1)$, so we can apply Z_{n-1} there. The remaining subspaces are the *crossing subspaces*, whose set is

$$\mathcal{X}_H = \{U \leq \mathbb{F}_q^n : U \not\leq H\}.$$

Let f_0 and f_1 be the restrictions of f to the subspaces inside H and to $\mathcal{X}_H$, respectively. Ordering both rows and columns with the inside subspaces first gives

$$\mathsf{Z}_n = \begin{pmatrix} \mathsf{Z}_{n-1} & 0 \\ C & \mathsf{A} \end{pmatrix}, \qquad \mathsf{Z}_n \begin{pmatrix} f_0 \\ f_1 \end{pmatrix} = \begin{pmatrix} \mathsf{Z}_{n-1} f_0 \\ C f_0 + \mathsf{A} f_1 \end{pmatrix}. \tag{2}$$

The zero block appears because a crossing subspace cannot be contained in an inside subspace. The block C sums contributions from inside subspaces to crossing subspaces, while A sums over containments between crossing subspaces.

For a crossing subspace W, the inside subspaces contained in W are exactly the subspaces of $W \cap H$. Therefore

$$(C f_0)(W) = \sum_{\substack{U \leq W \\ U \leq H}} f(U) = (\mathsf{Z}_{n-1} f_0)(W \cap H).$$

This value is already available from the transform inside H, so only $\mathsf{A} f_1$ remains to be computed.

Splitting the crossing set. One may try partitioning $\mathcal{X}_H$ into induced copies of smaller sets $\mathrm{Gr}_q(i)$, so that a zeta transform can be applied to each. However, containments between these sets would still make their transforms depend on each other. We therefore relax the induced-copy requirement and allow each crossing subspace to contribute its weight to several copies of smaller sets $\mathrm{Gr}_q(i)$. In each copy, sum the contributions to the same subspace and apply its zeta transform.

[3] The set containing the zero subspace has a containment relation with every subspace in the other set.

Then recover the crossing outputs from all the transformed copies. In matrix form, this asks for an invertible map $\mathcal{F}$ satisfying

$$\underbrace{\mathcal{F}(\mathsf{A}f_1)}_{\text{copied outputs}} = \operatorname{diag}(\mathsf{Z}_{i_1}, \dots, \mathsf{Z}_{i_m}) \underbrace{\mathcal{F}f_1}_{\text{copied inputs}}, \qquad i_j < n.$$

Summing contributions at the same subspace may lose information, so we must choose the contributions so that $\mathcal{F}$ is reversible. Fourier transformation is a natural choice: it is reversible and can be computed quickly by FFTs.

Choosing the smaller copies. We use one copy of $\mathrm{Gr}_q(n-1)$ and $q^{n-1}-1$ copies of $\mathrm{Gr}_q(n-2)$,[4] labeling them by the frequencies $\lambda \in \mathbb{F}_q^{n-1}$. We regard each vector λ as the linear functional $\lambda(x) = \lambda^{\mathsf{T}} x$, so its copy consists of the subspaces of $\ker \lambda$. This kernel has dimension $n-1$ when $\lambda = 0$ and $n-2$ otherwise. Each frequency labels a separate copy, even when two kernels coincide.

Choosing the Fourier transform. At frequency λ, applying Fourier transformation computes the contribution of f_1 to the corresponding copy. It remains to determine which specific subspace in that copy receives each contribution, while retaining the containment sums. To make this choice, represent each crossing subspace by two coordinates: its direction identifies the subspace inside a copy, and Fourier transformation of its translation determines the contributions to the copies.

Lemma 1 (Crossing subspaces)**.** *Every subspace $U \in \mathcal{X}_H$ has a unique pair $(B, x+B)$ such that*

$$U = U(B, x+B) := (B \times \{0\}) + \left\langle \begin{pmatrix} x \\ 1 \end{pmatrix} \right\rangle, \qquad B \le \mathbb{F}_q^{n-1}, \quad x + B \in \mathbb{F}_q^{n-1}/B. \tag{3}$$

Here $B \times \{0\} = U \cap H$ and $\dim U = \dim B + 1$. For $U_i = U(B_i, x_i + B_i)$, $i = 1, 2$,

$$U_1 \le U_2 \quad \Longleftrightarrow \quad B_1 \le B_2 \ \text{ and } \ x_1 - x_2 \in B_2.$$

The condition $B_1 \le B_2$ is exactly what we need to apply a smaller zeta transform inside each frequency copy. Keep B as the subspace coordinate and apply Fourier transformation to $x + B$ to remove the second condition from the sum while keeping $B_1 \le B_2$. For complex weights, use the additive character ψ from section 2.1. Each $U(B, x+B)$ contributes $f_1(U(B, x+B))\psi(\lambda(x))$ at subspace B in copy λ, whenever $B \le \ker \lambda$. Summing these contributions gives

$$\widehat{f_\lambda}(B) = \sum_{x + B \in \mathbb{F}_q^{n-1}/B} f_1\big(U(B, x+B)\big)\psi(\lambda(x)), \qquad B \le \ker \lambda. \tag{4}$$

Figure 1 illustrates this change of coordinates for $\mathbb{F}_2^3$.

Fourier inversion recovers each table on $\mathbb{F}_q^{n-1}/B$, so all the copies together retain the crossing weights. Let F_B denote the Fourier transform in (4) for fixed B. If P reorders the combined output indices (B, λ) as (λ, B), then lemma 2 gives

$$\mathcal{F} = P \left(\bigoplus_{B \le \mathbb{F}_q^{n-1}} F_B \right),$$

$$\mathsf{A} = \mathcal{F}^{-1} \operatorname{diag}\left(\mathsf{Z}_{n-1}, \underbrace{\mathsf{Z}_{n-2}, \dots, \mathsf{Z}_{n-2}}_{q^{n-1}-1 \text{ copies}} \right) \mathcal{F}.$$

[4] Their combined size is $N_q(n-1) + (q^{n-1}-1)N_q(n-2) = |\mathcal{X}_H|$, by (11) with cutoff $k = n$. For $n = 1$, only the copy labeled by $\lambda = 0$ occurs.

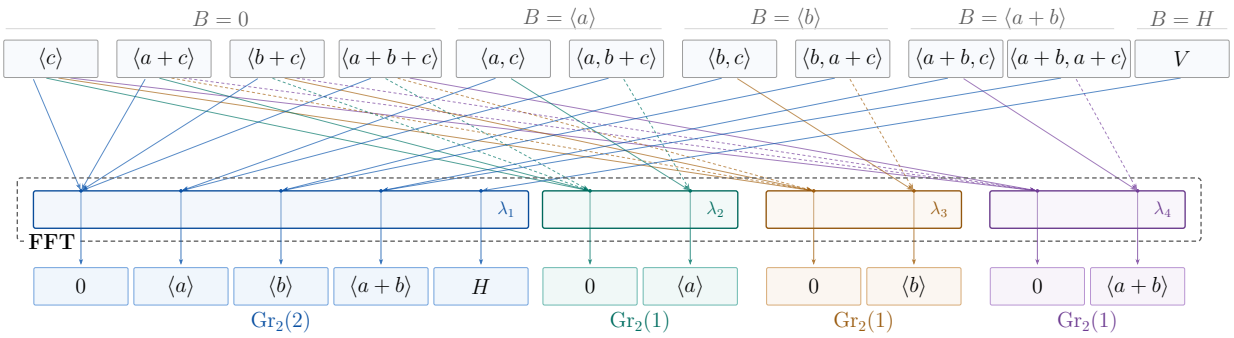


Figure 1: Fourier decomposition in $\mathbb{F}_2^3 = \langle a, b, c \rangle$, with $H = \langle a, b \rangle$. The top row groups crossing subspaces $U = B + \langle c + x \rangle$ by $B = U \cap H$, and Fourier transforms on H/B combine their weights with signs $(-1)^{\lambda(x)}$, shown by solid $(+1)$ and dashed (-1) arrows. Regrouping the coefficients by frequency λ gives the bottom blocks, indexed by $B \leq \ker \lambda$ with their induced containment order: within each block, every subspace is drawn to the left of those that strictly contain it.

These equations give a reversible map with the required containment sums in each copy. Algorithm 1 summarizes this recursion. A chosen basis identifies each kernel with its coordinate space.

Algorithm 1: The full zeta transform: $\mathrm{Sum}(n, f) = \mathsf{Z}_n f$

Input: A table f on $\mathrm{Gr}_q(n)$.
Output: The table $\mathsf{Z}_n f$.

if $n = 0$ **then return** f
Split f into f_0 on the subspaces inside H and f_1 on $\mathcal{X}_H$
$z_0 \leftarrow \mathrm{Sum}(n-1, f_0)$
$(\widehat{f}_\lambda)_{\lambda \in \mathbb{F}_q^{n-1}} \leftarrow \mathcal{F} f_1$
foreach $\lambda \in \mathbb{F}_q^{n-1}$ **do**
  $\widehat{u}_\lambda \leftarrow \mathrm{Sum}(\dim \ker \lambda, \widehat{f}_\lambda)$
end
$u \leftarrow \mathcal{F}^{-1}\left((\widehat{u}_\lambda)_{\lambda \in \mathbb{F}_q^{n-1}}\right)$
return $(\mathsf{Z}_n f)(W) = \begin{cases} z_0(W), & W \leq H, \\ z_0(W \cap H) + u(W), & W \in \mathcal{X}_H. \end{cases}$

Dimension cutoffs and running time. We now extend the recursion to $\mathsf{Z}_{n,k}$, which sums over subspaces of dimension at most k. The hyperplane partition still works: inside H the cutoff stays k, while for a crossing subspace $U(B, x+B)$, the condition $\dim U \leq k$ becomes $\dim B \leq k-1$. Thus the recursive transforms are

$$\mathsf{Z}_{n-1,k} f_0 \quad \text{and} \quad \mathsf{Z}_{\dim \ker \lambda, k-1} \widehat{f}_\lambda \quad (\lambda \in \mathbb{F}_q^{n-1}).$$

To bound the total work, first observe that the copying has not increased the number of entries. For each B, there are equally many cosets $x + B$ and frequencies with $\lambda(B) = 0$. The smaller transforms therefore have $M_q(n, k)$ entries in total, as expressed by (11), and the Fourier steps cost $O_q(n M_q(n, k))$.

We prove a total bound proportional to $nkM_q(n,k)$ by induction. The saving comes from the calls with cutoff $k-1$, which contain at least half the entries. For these entries, the factor nk decreases by at least $nk-(n-1)(k-1)\geq n$, and for the others it does not increase. This decrease covers the Fourier work after choosing a sufficiently large constant. Thus the total cost is $O_q(nkM_q(n,k))=O_q(M_q(n,k)\log M_q(n,k))$, as detailed in section 3.

Inversion and incidence. To compute $\mathsf{Z}_{n,k}^{-1}$, recover f from the supplied output (z_0,z_1) by solving the block equations in (2), with the dimension cutoff, in order:

$$f_0=\mathsf{Z}_{n-1,k}^{-1}z_0,\qquad f_1=\mathsf{A}^{-1}\big(z_1-Cf_0\big). \tag{5}$$

The inverse of A uses the same Fourier maps with inverse smaller zeta transforms. To obtain sums over superspaces as well, we transpose these linear computations: each weighted contribution is sent in the reverse direction and contributions at each input are added. This gives $\mathsf{Z}_{n,k}^{\mathsf{T}}$ and its inverse with the same order of work.

For the incidence transform $\mathsf{I}_n^{a,b}$, retain only a-dimensional inputs and b-dimensional outputs in the same hyperplane partition. Let $\mathsf{A}_{a-1,b-1}$ sum over crossing inputs with directions of dimension $a-1$ and crossing outputs with directions of dimension $b-1$. For inside and crossing inputs f_0,f_1, the output $z=(z_0,z_1)$ satisfies

$$\begin{aligned} z_0&=\mathsf{I}_{n-1}^{a,b}f_0,\\ z_1\big(U(B,x+B)\big)&=(\mathsf{I}_{n-1}^{a,b-1}f_0)(B)+(\mathsf{A}_{a-1,b-1}f_1)(B,x+B).\end{aligned}$$

Fourier transformation replaces $\mathsf{A}_{a-1,b-1}$ by $\mathsf{I}_{\dim\ker\lambda}^{a-1,b-1}$ at each frequency. The inside input f_0 is used in two calls, so section B accounts separately for this repetition and proves the $O_q(S\log S)$ bound for $S=|\operatorname{Gr}_q(n,a)|+|\operatorname{Gr}_q(n,b)|$. To extend the argument beyond complex weights, we must supply Fourier character values that may not lie in $\mathcal{R}$. Section 3.4 represents them using a constant number of elements of $\mathcal{R}$, so the same recursion runs at constant overhead. Its outputs are the required sums in $\mathcal{R}$, and its only divisions are by powers of p, as permitted in theorem 1.

1.2.2 Sums on Grassmann graphs

To compute Kf by two containment transforms, we will express each weight $K(U,W)$ as a sum over subspaces of $U\cap W$. Suppose $k\leq n/2$. An upper transform of f on $\operatorname{Gr}_q(n,k)$ gives $H(L)=\sum_{W\supseteq L,\,\dim W=k}f(W)$. Summing over the j-spaces $L\leq U$ counts each W once for every j-space in $U\cap W$:

$$\sum_{\substack{L\leq U\\ \dim L=j}}H(L)=\sum_{W\in\operatorname{Gr}_q(n,k)}|\operatorname{Gr}_q(\dim(U\cap W),j)|f(W).$$

The matrix $(|\operatorname{Gr}_q(t,j)|)_{j,t=0}^{k}$ is triangular with diagonal one. To apply K, choose $c_0,\dots,c_k$ by back-substitution so that $a_{k-t}=\sum_{j=0}^{t}|\operatorname{Gr}_q(t,j)|c_j$. Then

$$(Kf)(U)=\sum_{L\leq U}c_{\dim L}H(L).$$

Thus we apply K by an upper transform, multiplication of each $H(L)$ by $c_{\dim L}$, and a lower transform. For $k>n/2$, taking annihilators preserves distances and reduces to the preceding case.

Over $\mathbb{Q}$ or $\mathbb{C}$, interpolate P_j with $P_j(\theta_i)=\mathbf{1}[i=j]$ at the distinct eigenvalues of D_1. Then $P_j(D_1)f$ is the θ_j-component. Over fields with invertible p, symmetry makes K^{-1} an intersection kernel whenever K is nonsingular. Its $d+1$ coefficients follow from a linear system of that size. Appendix C proves these consequences.

1.2.3 Counting subsets by exact span

Given vectors $v_1, \ldots, v_\ell \in \mathbb{F}_q^n$, let $C(U)$ count the subsets of indices whose vectors span U. Weight each subspace by the number of input vectors spanning it—only the zero subspace and lines can have nonzero weight. A zeta transform gives $a(U) = |\{i : v_i \in U\}|$. Every subset using only these vectors has a unique span contained in U, so

$$2^{a(U)} = \sum_{V \leq U} C(V).$$

One Möbius transform therefore recovers all $C(U)$, with the running time given in theorem 5.

1.2.4 Graph coloring

To decide which classes can share a color across a cut (X, Y) of rank r, factor its adjacency matrix to give labels $p_x, q_y \in \mathbb{F}_2^r$ with $xy \in E$ exactly when $p_x^\mathsf{T} q_y = 1$. Two color classes on opposite sides can share a color exactly when the spans of their labels are orthogonal. Call these spans the classes' profiles.

First consider one coloring of each child of a decomposition node, with child vertex sets X_1, X_2. A parent coloring pairs some classes from the left child with classes from the right child, leaving unpaired classes separate. Each pair must have no edge between its two classes, a condition determined by their profiles. Let π_i restrict a child's neighborhood vectors to vertices outside $X_1 \cup X_2$. A paired class with child profiles U, V has parent profile $\pi_1(U) + \pi_2(V)$, while unpaired classes have profiles $\pi_1(U)$ and $\pi_2(V)$, respectively. Every parent color partition determines its child partitions and this partial matching uniquely.

We now compute all possible parent histograms (counts of classes with each profile) from the child histograms. Write $L = N_2(r)$. Recording a multiplicity for every pair of profiles would require up to L^2 counts. Instead, fix a left histogram and process its classes one at a time. Each class either stays separate or takes an available compatible right partner. We need only the numbers of unused right classes and the histogram of parent classes already formed. These are at most $2L$ counts, each at most the number of vertices N. There are $N^{O(L)}$ states and left histograms, at most N steps, and at most $L + 1$ choices per step, giving $N^{O(L)}$ total work. After the last left class, the remaining right classes stay separate. Choosing a partner with profile V has weight equal to the number of unused right classes of that profile, so the same recurrence counts partitions.

At the root, let a_t count the partitions into t nonempty independent classes. Assigning distinct colors to these classes gives the chromatic polynomial

$$P_G(z) = \sum_{t=0}^{N} a_t (z)_t, \qquad (z)_t = z(z-1)\cdots(z-t+1), \quad (z)_0 = 1.$$

The chromatic number is the least t with $a_t > 0$. Section 6 proves the recurrence and its counting weights.

For the fixed-palette lower bound, arrange k^2 Boolean variables in k rows. Give an assignment $u \in \mathbb{F}_2^k$ to row j the label $(e_j, u) \in \mathbb{F}_2^{2k}$, and an assignment $v \in \mathbb{F}_2^k$ to column h the label (v, e_h). Orthogonality is precisely the consistency condition $u_h = v_j$. Selecting at least one assignment in each row and column therefore forces one matrix of assignments, shared by all clause copies. Odd cycles enforce these nonempty choices and have exactly two extensions for each forced singleton selection. Constant-size clause gadgets give sixteen extensions for every satisfied clause. The same graph thus proves both the decision and exact counting bounds, with cut-rank $O(k)$.

For unrestricted palettes, we represent integer variables by numbers of colors. Use the 2^{d^2} subspaces $\{(x, A^{\mathsf{T}}x) : x \in \mathbb{F}_2^d\}$, one for each binary $d \times d$ matrix A, as color types, and let the number of colors of each type represent one variable. Orthogonality tests let a constraint use colors of a specified type, with total cut-rank $2d + O(1)$. A known ETH lower bound excludes $f(K)m^{o(K)}$ time for the source problem with K counters and domain size m [FGL+19, Lemma 3.1]. Choosing $d = \lceil \sqrt{\log_2 K} \rceil$ provides enough types and gives $r = 2\sqrt{\log_2 K} + O(1)$, yielding the coefficient $1/4$ in the exponent. Keeping the explicit additive constant in this width bound gives the sharper exclusion in theorem 8. The two reductions are detailed in sections G and H.

The reductions give quantitative implications without assuming ETH. For example, a $T(N, r)$-time algorithm for three-colorability would decide a k^2-variable 3-CNF formula with m clauses in time

$$2^{O(k)} \operatorname{poly}(m) + T\big(2^{O(k)} \operatorname{poly}(m), O(k)\big).$$

ETH is used only to exclude the resulting SAT running times.

Related Work The covering-product identity of Björklund, Husfeldt, Kaski, and Koivisto [BHKK07] gives span convolution from subspace containment sums. The covering-pair algorithm of [KKW16] uses one operation per covering pair on semimodular lattices.

Srinivasan [Sri14, Theorems 1.4 and 2.5] uses translation characters to decompose the operator summing over immediate superspaces. For $B \le \ker \lambda$, his unnormalized character projection of $U(B, B)$ has coefficient $q^{\dim B}\psi(-\lambda(x))$ at $U(B, x + B)$. Dividing by q^{n-1} gives the inverse Fourier column in (7). We bound the cost of containment transforms on arbitrary tables, including cutoffs and two prescribed dimensions, accounting for Fourier coordinate changes and recursive calls over rings with invertible p.

For coloring, we use the subspace profiles of [BTV10, GHO13]. Our transform answers compatibility queries on supplied profile tables, whereas the coloring algorithm instead merges profile multiplicities. Ganian, Hliněný, and Obdržálek [GHO13, Theorems 4.1 and 4.7] enumerate matching multiplicities for pairs of profiles to compute the chromatic number and polynomial. Our recurrence processes the left classes one profile at a time.

Bergougnoux, Korhonen, and Nederlof [BKN23] ask for tight rank-width bounds for fixed-palette coloring and Chromatic Number. Their reduction encodes assignments by subspaces and forces consistency through a quota of common nonneighbors [BKN23, Lemma 4.9, full version]. We retain the assignment encoding and enforce consistency by testing agreement between row and column assignments. Odd cycles and fixed-size clause gadgets realize these choices in three-coloring with cut-rank $O(k)$ for k^2 SAT variables, and give a fixed number of extensions per satisfying assignment.

For unrestricted palettes, we use the monotone-counter constraints and selector of Fomin, Golovach, Lokshtanov, Saurabh, and Zehavi [FGL+19]. Our matrix register and type tests store K counters at cut-rank $2\sqrt{\log_2 K} + O(1)$, yielding the matching coefficient $1/4$.

2 Preliminaries and definitions

We interpret $\mathrm{Gr}_q(n, j) = \varnothing$ when $j < 0$ or $j > n$ and use empty tables for negative cutoffs. Choose a basis for each space and write vectors as columns. For subspaces, $U + W = \{u + w : u \in U, w \in W\}$, and $U \oplus W$ means $U \cap W = \{0\}$. For $B \le H = \mathbb{F}_q^d$, an affine flat $x + B$ is a coset of its direction B. The cosets form the vector space H/B, with addition $(x + B) + (y + B) = (x + y) + B$ and dimension $\dim H - \dim B$. Two cosets agree exactly when their representatives differ by an element of B. We identify $\lambda \in \mathbb{F}_q^d$ with the linear map $\lambda(x) = \lambda^{\mathsf{T}}x$ and write $\ker \lambda = \{x : \lambda(x) = 0\}$. For $B \le \mathbb{F}_q^d$, its *annihilator* is $B^\circ = \{\lambda \in \mathbb{F}_q^d : \lambda(B) = 0\}$, of dimension $d - \dim B$.

A *table* on X is a function $f : X \to \mathcal{R}$, viewed as a vector in $\mathcal{R}^X$. Matrix rows index outputs and columns index inputs. Ordering subspaces by increasing dimension makes the containment matrix triangular with diagonal one, so its inverse exists over every ring. Our fast algorithm additionally requires that $p = \operatorname{char} \mathbb{F}_q$ be invertible. We count ring additions, subtractions, and multiplications, and field divisions for linear systems. Constants in $O_q(\cdot)$ depend only on q. Appendices A and D give the subspace counts, the construction of coordinates for the recursive tables, and the bit costs.

2.1 The finite Fourier transform

We use the Fourier transform on each quotient H/B, with a nontrivial additive character $\psi : \mathbb{F}_q \to \mathbb{C}$. For $q = p^m$, one choice is $\psi(t) = \exp(2\pi i \operatorname{Tr}(t)/p)$, where $\operatorname{Tr}(t) = t + t^p + \cdots + t^{p^{m-1}} \in \mathbb{F}_p$ and the exponential uses an integer representative. For $B \le H = \mathbb{F}_q^d$ and $h_B : H/B \to \mathbb{C}$, Fourier transformation and its inverse are

$$\widehat{h}_\lambda(B) = \sum_{x+B \in H/B} h_B(x+B)\psi(\lambda(x)), \qquad \lambda \in B^\circ, \tag{6}$$

$$h_B(x+B) = q^{-(d-\dim B)} \sum_{\lambda \in B^\circ} \widehat{h}_\lambda(B)\psi(-\lambda(x)). \tag{7}$$

The condition $\lambda(B) = 0$ makes these formulas independent of coset representatives, and both index sets have $q^{d-\dim B}$ entries. Inversion follows from

$$\sum_{\lambda \in B^\circ} \psi(\lambda(z)) = \begin{cases} q^{d-\dim B}, & z \in B, \\ 0, & z \notin B. \end{cases}$$

For the second case, $\lambda \mapsto \lambda(z)$ maps onto $\mathbb{F}_q$. Translating the resulting nontrivial character sum multiplies it by a value different from one, so the sum vanishes. To compute the transforms, identify H/B with $\mathbb{F}_p^{m(d-\dim B)}$ and apply a length-p transform along each coordinate, taking $O_q((d+1)|H/B|)$ operations. Section 3.4 implements these formulas over the stated coefficient rings.

3 A hyperplane cut and an ordinary Fourier transform

To prove theorem 1, we extend the overview's decomposition to subspaces of dimension at most k, working over $\mathbb{C}$ until section 3.4.

3.1 The recursion with a dimension cutoff

Proof of lemma 1. With $H = \mathbb{F}_q^{n-1} \times \{0\}$, let $B \times \{0\} = U \cap H$. Scale a vector of U with nonzero last coordinate to $\binom{x}{1}$. Subtracting multiples of this vector leaves exactly $B \times \{0\}$, giving (3). Two choices of x differ by an element of B, so $x + B$ is unique. For $U_i = U(B_i, x_i + B_i)$, containment $U_1 \le U_2$ holds exactly when $B_1 \le B_2$ and $\binom{x_1}{1} \in U_2$. The latter condition is $x_1 - x_2 \in B_2$. □

We now compute $g = \mathsf{Z}_{n,k} f$ recursively, using $H = \mathbb{F}_q^{n-1} \times \{0\}$ and $U(B, x+B)$ from the overview. Split the input into $f_0(B) = f(B \times \{0\})$ and $(f_1)_B(x+B) = f(U(B, x+B))$, and index the output parts g_0, g_1 in the same way. To sum the crossing inputs, lemma 1 requires sums over contained cosets. For tables $h_B : \mathbb{F}_q^{n-1}/B \to \mathbb{C}$, define

$$(\mathsf{A}h)_D(x+D) = \sum_{B \le D} \sum_{y+B \subseteq x+D} h_B(y+B).$$

With cutoff k, inside subspaces have dimension at most k, and crossing subspaces have directions of dimension at most $k-1$. Thus

$$g_0 = \mathsf{Z}_{n-1,k} f_0, \qquad (g_1)_D(x+D) = g_0(D) + (\mathsf{A} f_1)_D(x+D). \tag{8}$$

The value $g_0(D)$ comes from $\mathsf{Z}_{n-1,k} f_0$. It remains to compute $\mathsf{A} f_1$, which we do by Fourier transformation.

3.2 Why Fourier transformation removes translations

To evaluate A, we apply (6) to each coset table. The following identity leaves only the subspace containment condition. We state it for $H = \mathbb{F}_q^d$ and tables $h_B : H/B \to \mathbb{C}$ with $\dim B \le \ell$.

Lemma 2 (Affine Fourier identity)**.** *For every direction D of dimension at most ℓ and every $\lambda \in D^\circ$,*

$$\widehat{\mathsf{A} h}_\lambda(D) = \sum_{B \le D} \widehat{h}_\lambda(B). \tag{9}$$

Proof. For each $B \le D$, every B-coset lies in exactly one D-coset, and λ is constant on that D-coset. Hence the total contribution from direction B is

$$\sum_{x+D} \sum_{y+B \subseteq x+D} h_B(y+B)\psi(\lambda(x)) = \sum_{y+B} h_B(y+B)\psi(\lambda(y)) = \widehat{h}_\lambda(B).$$

Summing over B proves the formula. □

For each frequency λ, the identity requires a zeta transform on the subspaces of $\ker \lambda$. Let $\mathcal{F}$ apply (6) for each B and group the resulting entries by frequency. Order frequencies with 0 first, and order each table $(\widehat{h}_\lambda(B))_{B \le \ker \lambda,\, \dim B \le \ell}$ by increasing dimension. The identity gives

$$\mathsf{A} = \mathcal{F}^{-1} \operatorname{diag}\left(\mathsf{Z}_{d,\ell}, \underbrace{\mathsf{Z}_{d-1,\ell}, \dots, \mathsf{Z}_{d-1,\ell}}_{q^d - 1 \text{ blocks}} \right) \mathcal{F}. \tag{10}$$

A chosen basis identifies each kernel with its coordinate space, giving one block per frequency even when kernels coincide. For $d = 0$ there is only the first block.

Use $d = n-1$ and $\ell = k-1$ in (10) to compute $\mathsf{A} f_1$, then add $g_0(D)$ as in (8). In each call, replace k by $\min(k, n)$ and return the input if $k = 0$. All other calls decrease n. For $\mathsf{Z}_{n,k}^{-1}$, apply (5) to $g = (g_0, g_1)$: Cf_0 is the known value $g_0(D)$ at each crossing output $U(D, x+D)$. Invert (10) by using inverse smaller zeta transforms.

3.3 The work follows the number of entries

For each $B \le \mathbb{F}_q^{n-1}$, the $q^{n-1-\dim B}$ cosets correspond to equally many frequencies with $\lambda(B) = 0$. Thus the smaller table lengths satisfy the Goldman–Rota recurrence with a cutoff [Sri14]:

$$M_q(n,k) = M_q(n-1,k) + M_q(n-1,k-1) + (q^{n-1}-1) M_q(n-2,k-1), \qquad n \ge 2. \tag{11}$$

By section 2.1, all local Fourier transforms together take $O_q(n M_q(n,k))$ operations in a call with parameters (n,k).

The case $k = 0$ uses no arithmetic. For $k \geq 1$, give each entry a budget proportional to nk. Only $\mathsf{Z}_{n-1,k} f_0$ keeps cutoff k, whereas the transforms in (10) have cutoff $k-1$. At least half the entries belong to these latter transforms, because

$$M_q(n,k) - 2M_q(n-1,k) = (q^{n-1}-1)M_q(n-2,k-1) - |\operatorname{Gr}_q(n-1,k)| \geq 0.$$

For $1 \leq k < n$, use $|\operatorname{Gr}_q(n-1,k)| = (q^{n-1}-1)|\operatorname{Gr}_q(n-2,k-1)|/(q^k-1)$. For $k = n$ the last count vanishes, and $n = 1$ is direct. For each such entry, the budget decreases by at least $nk - (n-1)(k-1) = n + k - 1 \geq n$. All other entries have nonincreasing budgets, leaving at least $nM_q(n,k)/2$ for the local Fourier work. Induction gives the bound for the transform and its inverse. Clipping each cutoff to its ambient dimension only reduces the budget. For $1 \leq k \leq n/2$, (24) gives $\log_q M_q(n,k) \geq k(n-k) \geq nk/2$. For $k > n/2$, the middle-dimensional term gives $\log_q M_q(n,k) \geq \lfloor n^2/4 \rfloor = \Omega(nk)$.

To obtain sums over superspaces, apply the transposition principle [BLS03] to the circuit for $\mathsf{Z}_{n,k}$. This computes $\mathsf{Z}_{n,k}^{\mathsf{T}}$ with the same asymptotic cost. Attach a zero accumulator a_v to each circuit wire v. Add each new input to the accumulator of its corresponding output wire, so shared output wires receive the sum of these inputs. Traverse gates backwards:

$$z = x + y: \quad a_x \leftarrow a_x + a_z,\ a_y \leftarrow a_y + a_z, \qquad z = cx: \quad a_x \leftarrow a_x + ca_z.$$

The accumulators on the original input wires give the transpose. Applying the same procedure to the inverse circuit computes $(\mathsf{Z}_{n,k}^{\mathsf{T}})^{-1}$, preserving the cutoff.

3.4 Coefficient rings

To run the recursion over a commutative ring $\mathcal{R}$ with invertible p, we must supply the Fourier character values, which need not lie in $\mathcal{R}$. For this, adjoin a formal variable satisfying $1 + t + \cdots + t^{p-1} = 0$. Thus we work in

$$\mathcal{R}[t]/(1 + t + \cdots + t^{p-1}), \qquad \zeta = t \bmod (1 + t + \cdots + t^{p-1}), \qquad \psi(x) = \zeta^{\operatorname{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(x)}.$$

Each element is a polynomial reduced by this monic relation, hence has a unique expansion in $1, t, \ldots, t^{p-2}$. Thus $\mathcal{R}$ embeds as the constants and each extension operation costs $O_q(1)$ operations in $\mathcal{R}$. Also $\zeta^p = 1$ and $\sum_{j=0}^{p-1} \zeta^{aj} = 0$ for $a \not\equiv 0 \pmod p$. These identities prove Fourier orthogonality even when $\mathcal{R}$ has zero divisors, and all divisions are by powers of p. Since the outputs equal the original containment sums, all nonconstant coefficients vanish, so extracting the constant coefficient gives the answer in $\mathcal{R}$. This proves the coefficient statements without assuming roots of unity in $\mathcal{R}$ or dividing by subspace counts.

4 Incidence between two dimensions

Computing $\mathsf{I}_n^{a,b}$ via the truncated zeta transform $\mathsf{Z}_{n,b}$ would require $M_q(n,b)$ entries, which can be far more than $S := |\operatorname{Gr}_q(n,a)| + |\operatorname{Gr}_q(n,b)|$. To obtain the bound in theorem 2, which depends only on S, we retain only the two required dimensions in the recursion. Let $f : \operatorname{Gr}_q(n,a) \to \mathbb{C}$ and $g = \mathsf{I}_n^{a,b} f$, so g is indexed by $\operatorname{Gr}_q(n,b)$. For $0 < a < b < n$, cut at $H = \mathbb{F}_q^{n-1} \times \{0\}$. Use f_0, f_1, g_0, g_1 as in (8): the first parts are indexed by subspaces in H, and the second by pairs $(B, x+B)$ representing $U(B, x+B)$. The dimensions in f_0, g_0 are a, b, and the directions in f_1, g_1 have dimensions $a-1, b-1$.

Consequently

$$g_0(D) = (\mathsf{I}_{n-1}^{a,b} f_0)(D),$$
$$(g_1)_D(x+D) = (\mathsf{I}_{n-1}^{a,b-1} f_0)(D) + \sum_{\substack{y+B \subseteq x+D \\ \dim B = a-1}} (f_1)_B(y+B). \tag{12}$$

For g_0, only inputs inside the hyperplane can contribute. For a crossing output with direction D, the inside inputs must lie in D, explaining the second inside call with output dimension $b-1$. The affine term separates by frequency as in lemma 2, now into incidence between dimensions $a-1, b-1$ in $\ker \lambda$.

Thus the algorithm Fourier-transforms each crossing input table $(f_1)_B$, recursively applies $\mathsf{I}_{\dim \ker \lambda}^{a-1,b-1}$ at each frequency, and uses (7), dividing by q^{n-b}, to recover the affine sum. Add the second inside result as in (12). For the boundary cases, return f if $a = b$, copy $f(0)$ to every output if $a = 0$, and return $\sum_U f(U)$ if $b = n$.

Paying for the extra call. Equation (12) applies both $\mathsf{I}_{n-1}^{a,b}$ and $\mathsf{I}_{n-1}^{a,b-1}$ to f_0, so we must pay for using the same input twice. Let $T(n,a,b)$ be the arithmetic cost of $\mathsf{I}_n^{a,b}$. Input and output Fourier tables together have at most S entries, so

$$T(n,a,b) \le T(n-1,a,b) + T(n-1,a,b-1) + T(n-1,a-1,b-1)$$
$$+ (q^{n-1}-1)T(n-2,a-1,b-1) + O_q(nS). \tag{13}$$

The boundary cases cost $O(S)$. In every other call, all frequency blocks have valid input and output dimensions, so listing frequencies is also charged to their entries.

We pay for the extra call with a budget proportional to $n \min(a, n-b)$ per entry. The input dimension a and output codimension $n-b$ both decrease in the many nonzero-frequency calls, and this decrease pays for the repeated inside call. The resulting bound is $O_q(n \min(a,n-b)S) = O_q(S \log S)$ for the interior cases. Appendix B gives the induction. The ring extension in section 3.4 and circuit transposition then give the full statement of theorem 2.

5 Convolution, exact spans, and intersection kernels

5.1 Subspace convolution

For sums over pairs with $U \cap V = W$, use the upper transform $\mathsf{Z}_n^{\mathsf{T}}$ in place of Z_n in the span-convolution argument from the introduction. This takes $O_q(N_q(n) \log N_q(n))$ arithmetic operations. For direct sums $U \oplus V = W$, apply span convolution with inputs $f(U) z^{\dim U}$ and $g(V) z^{\dim V}$, and extract the coefficient of $z^{\dim W}$. Among pairs with $U+V = W$, the identity $\dim U + \dim V = \dim W + \dim(U \cap V)$ shows that this coefficient keeps exactly the pairs with $U \cap V = \{0\}$. Coefficientwise transforms and schoolbook polynomial multiplication give $O_q(n^3 N_q(n))$ arithmetic operations. This is the ranked-transform method of [BHKK07], applied to subspaces.

5.2 Counting subsets by their exact span

Möbius inversion also counts subsets with each possible span.

Theorem 5 (Counting exact spans)**.** *Given $v_1, \dots, v_\ell \in \mathbb{F}_q^n$, all numbers*

$$C(U) = \#\{A \subseteq [\ell] : \operatorname{span}\{v_i : i \in A\} = U\}, \qquad U \in \operatorname{Gr}_q(n),$$

can be computed in $O_q(\ell + N_q(n) \log N_q(n))$ arithmetic operations over $\mathbb{Q}$. The vectors may repeat and may be zero.

Proof. For each subspace B, let $t(B)$ count the indices i with $\langle v_i \rangle = B$. This table is supported on the zero subspace and the lines. Its zeta transform counts the input vectors in each subspace: $a(U) = (\mathsf{Z}_n t)(U) = |\{i : v_i \in U\}|$. There are $2^{a(U)}$ subsets of these indices. Partitioning them by their span gives

$$2^{a(U)} = \sum_{V \le U} C(V), \qquad C = \mathsf{Z}_n^{-1}\big(U \mapsto 2^{a(U)}\big). \tag{14}$$

Precompute $2^0, \dots, 2^\ell$ using ℓ doublings and index this list by $a(U)$. Building t takes $O(\ell)$ coefficient additions once the vectors are indexed by their spans. The two transforms then give the bound. Repeated vectors are distinct indexed choices, and $C(\{0\}) = 2^{t(\{0\})}$ includes the empty subset. □

Normalizing and indexing the vectors takes $\ell \operatorname{poly}(n)$ operations over $\mathbb{F}_q$ for fixed q, in addition to $N_q(n) \operatorname{poly}(n)$ work for the transform indices. Since the inversion input has at most $\ell + 1$ bits per entry, corollary 2 gives $(\ell + N_q(n)) \operatorname{poly}(n, \ell)$ total bit operations. For a uniformly chosen subset of the input indices, $C(U)/2^\ell$ is the probability that its span is exactly U.

5.3 Intersection kernels

To compute $(Kf)(U)$, we express the weight of $f(W)$ as a sum over subspaces of $U \cap W$, using the notation and coefficient assumption of theorem 3.

Intersection kernels are exactly the matrices satisfying $K(gU, gW) = K(U, W)$ for every invertible linear map $g \in \mathrm{GL}(n, q)$: extending bases sends any pair to any other pair with the same intersection dimension. To verify the distance formula, each edge changes the intersection dimension with W by at most one, and replacing the basis vectors of U outside $U \cap W$ by those of W attains the bound.

A double-counting identity. The cases $k = 0, n$ have one entry and take $O(1)$ operations. Suppose $1 \le k \le n/2$ and extend f by zero to dimensions below k. The upper transform gives $H(L) = (\mathsf{Z}_{n,k}^{\mathsf{T}} f)(L) = \sum_{W \in V \colon L \le W} f(W)$ for every L with $\dim L \le k$. For $U \in V$ and $0 \le j \le k$, summing these values over the j-dimensional subspaces of U gives

$$m_j(U) = \sum_{\substack{L \le U \\ \dim L = j}} H(L) = \sum_{W \in V} |\operatorname{Gr}_q(\dim(U \cap W), j)| f(W). \tag{15}$$

To assign weight a_{k-t} to intersection dimension t, choose $c_0, \dots, c_k$ by

$$a_{k-t} = \sum_{j=0}^{t} |\operatorname{Gr}_q(t, j)| c_j, \qquad c_t = a_{k-t} - \sum_{j<t} |\operatorname{Gr}_q(t, j)| c_j. \tag{16}$$

Back-substitution takes $O(k^2)$ operations. Now weight each upper-transform value $H(L)$ by $c_{\dim L}$ and apply one lower transform. At $U \in V$, its output is

$$\sum_{L \le U} c_{\dim L} H(L) = \sum_{j=0}^{k} c_j m_j(U) = (Kf)(U). \tag{17}$$

The subspace counts in (16) can be generated by (25) without division, so this works over every stated ring. Since $M_q(n, k) = \Theta_q(N)$ and $\log N = \Theta_q(nk)$, the two transforms and the $O(k^2)$ additional work take $O_q(N \log N)$ operations.

For $k > n/2$, annihilators preserve distance, since $\dim(U^\circ \cap W^\circ) = n - 2k + \dim(U \cap W)$. This proves theorem 3 for all k.

A dimension cutoff preserves intervals between retained subspaces, so the earlier covering-pair transforms [KKW16] still use one addition per covering pair. For $k \le n/2$, there are $\sum_{j=1}^{k} |\operatorname{Gr}_q(n,j)|[j]_q = \Theta_q(q^k N)$ such pairs: successive terms grow by at least q^2. Using these transforms in (17), and annihilators for $k > n/2$, gives $O_q(q^d N)$ operations.

6 Subspace profiles and graph coloring

To combine colorings across a cut, we record the span of the neighborhood vectors of each color class, following [BTV10, GHO13]. These subspace profiles let us test which classes can share a color.

Theorem 6 (Coloring on a supplied rank-decomposition)**.** *Given an N-vertex graph and a rank-decomposition of width at most r, its chromatic number, an optimal coloring, and its chromatic polynomial can be computed in*

$$N^{O(N_2(r))} = N^{2^{r^2/4+O(1)}}$$

bit operations.

Combining one pair of colorings. Suppress degree-two vertices of the decomposition and subdivide an edge to root it as a binary tree. For the vertex set X below a node, the neighborhood vectors into $V(G) \setminus X$ span

$$W_X = \operatorname{col}(A_G[X, V(G) \setminus X]^{\mathsf{T}}) \le \mathbb{F}_2^{V(G)\setminus X}, \qquad \dim W_X \le r.$$

A *color partition* is a partition into nonempty independent sets, with no names assigned to the classes. For a color partition of $G[X]$, the profile of a class C is

$$U_X(C) = \operatorname{span}\{A_G[x, V(G) \setminus X]^{\mathsf{T}} : x \in C\} \le W_X.$$

Consider one color partition on each child set X_1, X_2, and put $X = X_1 \dot\cup X_2$, $Y = V(G) \setminus X$, and $W_i = W_{X_i}$. We must decide which classes can share a color and find their parent profiles.

Restriction to Y gives linear maps $\pi_i : W_i \to W_X$. To test compatibility, define a bilinear form $\beta : W_1 \times W_2 \to \mathbb{F}_2$. For representations

$$u = \sum_{x \in X_1} a_x A_G[x, X_2 \cup Y]^{\mathsf{T}}, \qquad v = \sum_{y \in X_2} b_y A_G[y, X_1 \cup Y]^{\mathsf{T}},$$

set $\beta(u,v) = \sum_{x\in X_1, y\in X_2} a_x b_y A_G[x,y]$. Changing either representation adds a combination vanishing on the opposite child, so β is well-defined.

Lemma 3 (Compatibility and the parent profile)**.** *Classes with profiles $U \le W_1$ and $V \le W_2$ can share a color exactly when $\beta(U,V) = \{0\}$. Their union then has profile $\pi_1(U) + \pi_2(V)$. The separate classes have profiles $\pi_1(U)$ and $\pi_2(V)$, respectively.*

Proof. On the neighborhood vectors of vertices $x \in X_1$ and $y \in X_2$, the value of β is $A_G[x,y]$. Bilinearity makes its vanishing on $U \times V$ equivalent to the absence of all edges between the classes. Restricting their vectors to Y gives the stated parent profiles. □

Classes sharing a color form a compatible partial matching: each class has at most one partner. Conversely, such a matching gives a parent partition. Restricting it to X_1, X_2 recovers both child partitions and their matching, so the correspondence is a bijection.

Grouping partial colorings. Compatibility and parent profiles depend only on the child profiles. We therefore group color partitions by the histogram $h(U) = |\{C : U_X(C) = U\}|$, and store the number of color partitions realizing each histogram. Nonempty classes with zero profile contribute to $h(\{0\})$, so $\sum_U h(U)$ is the number of classes.

Fix a left histogram h_1. Rather than choose a multiplicity for every pair of profiles, as in [GHO13], process its classes one at a time, in a fixed order of profiles. Retain two histograms: $b(V)$ counts right classes of profile $V \le W_2$ still available, and $H(T)$ counts parent classes of profile $T \le W_X$ already formed. For every right histogram h_2, initialize $(b, H) = (h_2, 0)$ with the product of the two child counts.

Write δ_T for the histogram counting one class of profile T. When the next left class has profile U, either leave it separate or pair it with an available compatible right class:

$$(b, H) \longmapsto (b, H + \delta_{\pi_1(U)}), \qquad \text{with weight } 1, \tag{18}$$

$$(b, H) \longmapsto (b - \delta_V, H + \delta_{\pi_1(U)+\pi_2(V)}), \qquad \text{with weight } b(V), \tag{19}$$

where the second update requires $b(V) > 0$ and $\beta(U, V) = \{0\}$. Multiply each state's count by the transition weight and add it to the count of the resulting state. After all left classes have been processed, add each count to the parent histogram $H + \sum_{V \le W_2} b(V)\delta_{\pi_2(V)}$. Sum these contributions over all left histograms h_1.

To justify the weights, fix two concrete child partitions and order the left classes of each profile by their least vertex. The next left class is then fixed: it has one unmatched choice and exactly $b(V)$ choices of a right partner with profile V. Each such choice has the same new state, and later choices depend only on that state and the remaining left profiles. Thus counts can be combined even when they come from different right histograms. Every partial matching has a unique sequence of these choices. The bijection above therefore counts every parent partition once.

Initialization and output. At a leaf $\{v\}$, there is one partition, with one class of profile $\operatorname{span}\{A_G[v, V(G) \setminus \{v\}]^{\mathsf{T}}\}$. At the root, $W_{V(G)} = \{0\}$. The entry for $h(\{0\}) = t$ counts partitions into t nonempty independent classes. Denote it by a_t. Assigning distinct colors to the t classes of each partition gives the chromatic polynomial

$$P_G(z) = \sum_{t=0}^{N} a_t (z)_t. \tag{20}$$

Convert to ordinary coefficients by successively multiplying by $z - t$. The chromatic number is the least t with $a_t > 0$. Store one predecessor and pairing choice per positive state, and one contributing left histogram and terminal state per parent histogram. Backtracking recovers a right histogram and the pairing choices. After recursively recovering the child partitions, follow these choices using any available classes of the recorded profiles. Lemma 3 makes each pairing valid.

Running time in theorem 6. Put $L = N_2(r)$. Each profile space has at most L subspaces, and each histogram entry is at most N. There are at most $(N+1)^L$ left histograms. For each one, the merge has at most N steps, $(N+1)^{2L}$ states per step, and $L+1$ transitions per state. Over all nodes this gives $(N+1)^{3L} \operatorname{poly}(N, L)$ bit operations. Gaussian elimination, subspace enumeration, and precomputing restrictions, bilinear forms, compatibility tests, and profile sums cost $\operatorname{poly}(N, L)$. Pairs of partitions and partial matchings number $N^{O(N)}$, so all stored counts have $O(N \log N)$ bits, and transition weights are at most N. Conversion of (20) uses the same bit bound, so exact arithmetic costs a polynomial factor.

Since $\log L = O(L)$, the bound is $N^{O(L)}$, where $L = 2^{r^2/4+O(1)}$ by section A. □

Linear layouts.

Corollary 1. *On a supplied linear layout of width at most r, the three tasks in theorem 6 take $(N+1)^{N_2(r)}2^{O(r^2)}N^{O(1)}$ bit operations.*

Proof. On a linear layout, adding one vertex lets us update one histogram at a time. At a prefix X, let x be the next vertex and $R = V(G) \setminus (X \cup \{x\})$. On W_X, let ℓ evaluate the x-coordinate, let π restrict to R, and put $b = A_G[x, R]^{\mathsf{T}}$. Deleting the x-coordinate sends a histogram h to $(\pi_* h)(V) = \sum_{U:\pi(U)=V} h(U)$. The next vertex starts a new class or joins an existing one:

$$h \longmapsto \pi_* h + \delta_{\mathrm{span}(b)}, \tag{21}$$

$$h \longmapsto \pi_*(h - \delta_U) + \delta_{\pi(U)+\mathrm{span}(b)} \quad \text{if } h(U) > 0 \text{ and } \ell(U) = \{0\}. \tag{22}$$

Starting a new class has one choice, while joining a class of profile U has $h(U)$ choices. Starting with the empty histogram of count one, multiply by these weights and add counts at equal histograms. Removing the last vertex reverses exactly one choice, proving the counting invariant. With $L = N_2(r)$, there are at most $(N+1)^L$ histograms, $L+1$ choices per histogram, and $\mathrm{poly}(N, L)$ bit operations per choice. □

Compatibility queries on supplied profile tables. Labeled colorings across a binary rank-r cut combine exactly when corresponding profiles in $\mathbb{F}_2^r$ are orthogonal. Unused colors have profile $\{0\}$.

Proposition 1 (Fast coloring-completion queries)**.** *Let $c \geq 1$ and $S = N_2(r)^c$. Given vectors f, g indexed by c-tuples of subspaces of $\mathbb{F}_2^r$, with entries in a commutative ring in which 2 is invertible, all sums*

$$B_g(U_1, \ldots, U_c) = \sum_{\substack{V_i \leq U_i^\perp \\ 1 \leq i \leq c}} g(V_1, \ldots, V_c)$$

and $\sum_U f(U) B_g(U)$ take $O(S \log S)$ arithmetic operations. Here $U^\perp$ uses the usual dot product.

Proof. Apply Z_r in each argument of g, holding the other arguments fixed:

$$B_g(U_1, \ldots, U_c) = (\mathsf{Z}_r^{\otimes c} g)(U_1^\perp, \ldots, U_c^\perp). \tag{23}$$

These $cN_2(r)^{c-1}$ calls cost $O(cS \log N_2(r)) = O(S \log S)$, and the inner product costs $O(S)$. □

When f and g count labeled c-colorings of the two sides by their ordered profile tuples, the final sum counts labeled c-colorings of G. With H-bit integer entries, the bit cost over $\mathbb{Q}$ is $S\,\mathrm{poly}(c, r, H)$ by section D.

7 Experiments

Our algorithms give substantial speedups in all three applications (Table 1). The implementations, measurements, and scripts that generate the tables and figure are available in the companion repository [PW26].

Operation	Size	Speedup	Operation count ratio
Zeta transform Z_n	$N_q(n):\ 1.2 \cdot 10^6$–$6.7 \cdot 10^8$	1.5–18	0.45–4.1
Incidence $\mathsf{I}_n^{a,b}$	$S:\ 1.8 \cdot 10^6$–$8.2 \cdot 10^8$	4.5–190	4.3–170
Grassmann kernel Kf	$N:\ 2.6 \cdot 10^6$–$1.6 \cdot 10^8$	4.8–22	0.81–4.6

Table 1: Measured running time and operation count of the comparison algorithm divided by ours, for tests with at least 10^6 entries and measured running times for both algorithms. Values above one favor our algorithms.

We compare the zeta transform Z_n with the covering-pair algorithm [BHK+16, KKW16], and the incidence transform $\mathsf{I}_n^{a,b}$ with the faster of the covering-pair algorithm and Poisson summation [MP13, Theorem 10.1.16], using their measured running times. For operation counts, we compare our incidence transform with direct summation, the covering-pair algorithm, and Poisson summation, using the smallest of their counts. For Grassmann kernels, we tested $K(U, W) = 2^{-\dim(U,W)}$ against the distance-matrix recurrence [MOR03, Section 5.3], which computes $D_0 f, \ldots, D_d f$ before forming their weighted sum and uses incidence sums for each adjacency multiplication.

The speedups often exceed the operation count ratios, and our algorithms can have shorter measured running times despite using more operations. This may partly reflect how the implementations read memory: our algorithm stores corresponding entries of the frequency subproblems next to one another, so many additions and FFT steps scan contiguous rows, while the covering-pair algorithm fetches a subspace value through a stored index for each addition. For Grassmann kernels, Table 3 also reports measured running times with the covering-pair algorithm in place of our fast transforms in (17), showing how much of the speedup comes from this identity and how much from our fast transforms.

Figure 2 shows how the speedup for the zeta transform Z_n generally grows with table size and field size, though the covering-pair algorithm can have shorter measured running times on small tables. Poisson summation also has a shorter measured running time on the point–hyperplane incidence test. Appendix E gives the measurements and implementation details.

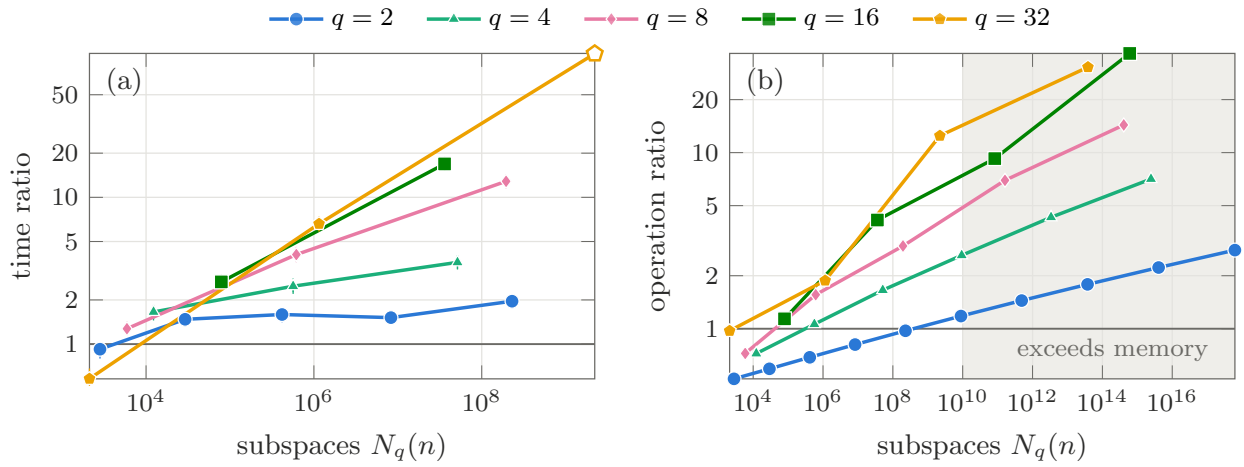


Figure 2: Measured running time and operation count of the covering-pair algorithm divided by those of our zeta transform Z_n. Higher ratios favor our algorithm. Shading marks instances on which the covering-pair algorithm exceeds the available memory. For the yellow hollow marker, we estimated its running time by processing a sample of covering pairs and scaling to all pairs.

A Counting subspaces

To count r-dimensional subspaces, with $0 \le r \le n$, we count their ordered bases: there are $\prod_{i=0}^{r-1}(q^n - q^i)$ ordered linearly independent r-tuples in $\mathbb{F}_q^n$. Dividing by the number of bases of each r-space gives

$$|\operatorname{Gr}_q(n,r)| = \prod_{i=0}^{r-1} \frac{q^n - q^i}{q^r - q^i} = q^{r(n-r)} \prod_{i=0}^{r-1} \frac{1 - q^{i-n}}{1 - q^{i-r}} = \Theta_q(q^{r(n-r)}). \tag{24}$$

The estimate is uniform in n, r: the last product is between 1 and $\prod_{i\ge1}(1-q^{-i})^{-1} < \infty$. Annihilators give $|\operatorname{Gr}_q(n,r)| = |\operatorname{Gr}_q(n,n-r)|$. We also write $[r]_q = |\operatorname{Gr}_q(r,1)| = (q^r-1)/(q-1)$ for the number of lines in an r-space. These integers can be computed without division, starting from $|\operatorname{Gr}_q(0,0)| = 1$, by

$$|\operatorname{Gr}_q(n,r)| = |\operatorname{Gr}_q(n-1,r)| + q^{n-r}|\operatorname{Gr}_q(n-1,r-1)|. \tag{25}$$

Indeed, the first term counts spaces in a fixed hyperplane, and each $(r-1)$-space in that hyperplane has q^{n-r} extensions outside it.

Completing the square in $r(n-r)$ gives $N_q(n) = \Theta_q(q^{\lfloor n^2/4 \rfloor})$. For dimensions at most $n/2$,

$$M_q(n,k) = \Theta_q(|\operatorname{Gr}_q(n,k)|) \qquad (0 \le k \le n/2), \tag{26}$$

because the exponent for dimension $k-j$ is smaller than for dimension k by $j(n-2k)+j^2 \ge j^2$, and $\sum_{j\ge0} q^{-j^2}$ is bounded. The covering-pair algorithm processes pairs $U < W$ whose dimensions differ by one. An r-dimensional subspace has $[n-r]_q$ immediate superspaces, one for each line in its quotient. Their total number is

$$\sum_{r=0}^{n-1} |\operatorname{Gr}_q(n,r)|[n-r]_q = \Theta_q\Big(q^{\lfloor (n+1)^2/4 \rfloor - 1}\Big) = \Theta_q(N_q(n) q^{n/2}). \tag{27}$$

Indeed, the summand has order $q^{r(n-r)+n-r-1} = q^{(r+1)(n-r)-1}$. Completing the square gives the displayed exponent, and terms farther from the maximum contribute a bounded sum of powers of the form q^{-j^2}, as above.

B Running time for incidence between two dimensions

The extra inside call in (13) repeats some entries. We pay for it using the decrease of the input dimension or output codimension in the frequency calls. Put $c = n - b$, $A = |\operatorname{Gr}_q(n,a)|$, $B = |\operatorname{Gr}_q(n,b)|$, and $S = A + B$. Write

$$I = |\operatorname{Gr}_q(n-1,a)| + |\operatorname{Gr}_q(n-1,b)|, \qquad J = |\operatorname{Gr}_q(n-1,a)| + |\operatorname{Gr}_q(n-1,b-1)|$$

for the lengths of the two inside calls. The fixed-dimension version of (11) shows that the frequency calls together have $S - I$ entries. All recursive calls therefore have $S + J \le 2S$ entries.

Assign a call the budget $\min(a,c)S$. Exchanging a, c swaps the first and third calls in (13), so assume $a \le c$. Every frequency call reduces this minimum by one. The first inside call does so exactly when $a = c$, while the second keeps it unchanged. Thus the budget decreases by

$$\min(a,c)S - \sum_{\text{recursive calls}} \min(a',c')S' = \begin{cases} S - aJ, & a = c, \\ S - I - aJ, & a < c, \end{cases} \quad \ge S/4. \tag{28}$$

To verify the inequality, use $|\operatorname{Gr}_q(n-1,j)|/|\operatorname{Gr}_q(n,j)| \le q^{-j}$. If $a = c$, annihilators give $J \le q^{-a}S$, so the decrease is at least $(1 - aq^{-a})S \ge S/2$. If $a < c$, then $a < b < n - a$, so $B \ge A$, and

$$S - I - aJ \ge \big(1 - (a+1)q^{-a}\big)A + \big(1 - q^{-b} - aq^{-c}\big)B \ge B/2 \ge S/4.$$

The last step uses $b, c \ge a + 1$, $(a+1)q^{-a} \le 1$, and $(a+1)q^{-(a+1)} \le 1/2$.

The boundary calls cost $O(S)$. Charge nonroot calls with $a = 0$ or $c = 0$ to their parents, using the bound $S + J \le 2S$. Since child dimensions decrease, multiplying (28) by n shows that a constant multiple of $n \min(a, c)S$ pays for the local $O_q(nS)$ work and all remaining calls. For an interior call, with $h = b - a$, (24) gives

$$n \min(a, c) \le 2\big(ac + h \max(a, c)\big) \le 2 \log_q S.$$

This proves $O_q(S \log S)$. Charging $\operatorname{poly}(n)$ per entry in the same argument also bounds index construction by $S \operatorname{poly}(n)$.

C Further applications on the Grassmannian

We derive further consequences of theorem 3, using the same notation.

Distance sums. For $k = 0, n$, the only distance sum is f. Otherwise first suppose $k \le n/2$ and reuse the table H from (15). To recover all distance sums, apply a lower transform to $H(L)\mathbf{1}[\dim L = j]$ for each $0 \le j \le k$. This computes all m_j in (15). For fixed U, let $h_t(U) = \sum_{W \in V : \dim(U \cap W) = t} f(W)$, $0 \le t \le k$. Grouping (15) by intersection dimension gives

$$\begin{aligned} m_j(U) &= \sum_{t=j}^{k} |\operatorname{Gr}_q(t, j)| h_t(U), \\ h_t(U) &= m_t(U) - \sum_{r=t+1}^{k} |\operatorname{Gr}_q(r, t)| h_r(U). \end{aligned} \tag{29}$$

Back-substitution starts with $h_k = m_k$ and recovers $h_{k-1}, \dots, h_0$, giving the desired outputs $D_i f = h_{k-i}$. The $k + 2$ transforms and $O(k^2 N)$ substitution work take $O_q((k+1)N \log N)$ operations. Annihilators preserve distance and handle $k > n/2$, so the full bound is $O_q((d+1)N \log N)$.

Spectral projections. To recover each eigenspace component of f, we construct a polynomial that is one on its eigenvalue and zero on the others. Over $\mathbb{Q}$ or $\mathbb{C}$, the real symmetric adjacency matrix D_1 has eigenvalues [Sri14, MOR03]

$$\theta_j = q[k]_q[n-k]_q - [j]_q[n-j+1]_q, \qquad 0 \le j \le d.$$

They are distinct because $\theta_j - \theta_{j+1} = q^j[n-2j]_q > 0$ for $j < d$. The polynomial and the θ_j-component f_j are

$$P_j(z) = \prod_{\substack{0 \le t \le d \\ t \ne j}} \frac{z - \theta_t}{\theta_j - \theta_t}, \qquad f_j = P_j(D_1)f.$$

Compute $f, D_1 f, \dots, D_1^d f$ by fast kernel multiplication, then combine them using the coefficients of P_j. The total cost is $O_q((d+1)N \log N)$, including the $O((d+1)^3 + (d+1)^2 N)$ interpolation work. For $d = 0$ the projection is the identity.

Invariant linear systems. Over a field of characteristic different from p, let $K(U,W) = a_{\dim(U,W)}$. To test invertibility and recover one inverse column, fix $E \in V$ and put $\mathcal{O}_j = \{U : \dim(U,E) = j\}$. Invertible linear maps fixing E are transitive on each $\mathcal{O}_j$, by extending bases, and commute with K. Thus K preserves the span of the indicators $b_j = \mathbf{1}_{\mathcal{O}_j}$. Choosing $U_i \in \mathcal{O}_i$ gives its matrix on this space:

$$B_{ij} = (Kb_j)(U_i), \qquad 0 \le i,j \le d. \tag{30}$$

If K is invertible, its inverse also preserves this space, so B is invertible. Conversely, if B is invertible, solve $Bc = (1,0,\dots,0)^{\mathsf{T}}$ and set $C(U,W) = c_{\dim(U,W)}$. Then $(KC)(U,E) = \mathbf{1}[U = E]$, and invariance gives every other column, proving $KC = I$. Constructing B, solving this small system, and applying $C = K^{-1}$ therefore test nonsingularity and solve $Kx = b$ in $O_q((d+1)N \log N)$ operations. No division by $|\mathcal{O}_j|$ is needed, even when eigenvalues of D_1 coincide in the coefficient field.

Distances to a specified family of subspaces. For a nonempty family $\mathcal{C} \subseteq V$, also called a constant-dimension code [KK08], compute $D_i\mathbf{1}_{\mathcal{C}}$ over $\mathbb{Q}$. These values count codewords at each distance, so

$$\delta_{\mathcal{C}}(U) = \min\{i : (D_i\mathbf{1}_{\mathcal{C}})(U) > 0\}$$

is the distance to the nearest codeword. The usual subspace distance is twice the graph distance, giving covering radius $2\max_{U\in V}\delta_{\mathcal{C}}(U)$. For $|\mathcal{C}| \ge 2$, its minimum code distance is twice the smallest $i > 0$ for which $(D_i\mathbf{1}_{\mathcal{C}})(U) > 0$ for some $U \in \mathcal{C}$. Summing counts over $\mathcal{C}$ gives ordered-pair distance distributions. Computing all distance sums gives these counts, but visits every subspace in V and need not be efficient for a sparse code or a single query.

Different input and output dimensions. With $\mathcal{R}$ as in theorem 1, let K have columns indexed by $\mathrm{Gr}_q(n,a)$ and rows by $\mathrm{Gr}_q(n,b)$, with $K(W,U)$ depending only on $\dim(U \cap W)$. For input $f : \mathrm{Gr}_q(n,a) \to \mathcal{R}$, the same double count gives, for $W \in \mathrm{Gr}_q(n,b)$ and $0 \le t \le \min(a,b)$,

$$\big(\mathsf{I}_n^{t,b}(\mathsf{I}_n^{t,a})^{\mathsf{T}} f\big)(W) = \sum_{U \in \mathrm{Gr}_q(n,a)} |\mathrm{Gr}_q(\dim(U\cap W), t)| f(U). \tag{31}$$

If $a, b > n/2$, take annihilators, which change the intersection dimension to $n - a - b + \dim(U \cap W)$. Now at least one of a, b is at most $n/2$, so each intermediate table is no larger than one of the input and output tables. Back-substitution in (29) recovers sums by intersection dimension, or applies any intersection kernel, in $S\,\mathrm{poly}(n)$ work, where $S = |\mathrm{Gr}_q(n,a)| + |\mathrm{Gr}_q(n,b)|$. For unions of dimensions, sum over dimension pairs and let S be the combined table length. On the table of all subspaces, one upper transform per input dimension and one weighted lower transform per output dimension give $O_q((n+1)^3 N_q(n))$ operations.

D Index construction and exact bit costs

Constructing the tables. To locate the entries used by each recursive transform, represent subspaces by reduced row-echelon bases and cosets by reduced representatives. Gaussian elimination constructs annihilators, kernels, quotients, and coordinate changes. For a call on $\mathbb{F}_q^n$ and a crossing direction $B \le \mathbb{F}_q^{n-1}$, enumerate (B,λ) with $\lambda \in B^\circ$. Express B in a chosen basis of $\ker\lambda$ to locate its entry in the transform for frequency λ. Group these entries by λ for the smaller transforms and by B for Fourier inversion. Each call has $q^{O(n^2)}$ records of $O(n^2)$ field elements, so elimination and sorting cost $\mathrm{poly}(n)$ per record. Equations (11) and (28) charge the recursive records, including repeated incidence inputs, giving table size times $\mathrm{poly}(n)$ construction work.

Exact integer and modular evaluation.

Corollary 2. *For fixed q, suppose a transform in theorems 1 and 2 has integer input of magnitude below 2^H and combined table length S. After index construction, its exact output takes*

$$O_q\big(S(H+n^2+1)^2 \log S\big)$$

bit operations. Reduction modulo any positive integer adds its reduction cost, and scaling handles rational inputs with a supplied common denominator.

Proof. We bound the outputs, then compute modulo a larger integer. For dimension gap j, the inverse containment coefficient is $(-1)^j q^{j(j-1)/2}$, as (25) verifies by interval inversion. Cutoffs preserve intervals. Thus all transform coefficients are integers of magnitude at most $q^{n(n-1)/2}$, and outputs have magnitude below $B = 2^H S q^{n(n-1)/2}$. Choose $L \in \{2B+1, 2B+2\}$ coprime to p. The transform applies over $\mathbb{Z}/L\mathbb{Z}$, and its centered residues uniquely recover the integer outputs because $L > 2B$. The modulus has $H + O_q(n^2+1)$ bits, so schoolbook modular arithmetic gives the bound. □

For fixed, explicitly represented finite coefficient fields, evaluate prime-field coordinates over the integers and reduce, even in characteristic p, though the arithmetic bound still requires invertible p. Span convolution has polynomial bit overhead, and exact-span counting uses $H = \ell + 1$. For rational kernel coefficients of bit length H, determinant bounds in (30) give inverse coefficients of polynomial bit length. Consequently kernel applications and solving take table size times $\mathrm{poly}(n, H)$ bit operations for integer entries or entries sharing a supplied common denominator of bounded bit length.

E Experimental details

The `experiments/` directory of the repository [PW26] contains the raw measurements, benchmark drivers, and analysis scripts. Its README gives the commands for reproducing the tables and figure from the measurements and for running the experiments again.

Implementation and measurements. We measured running times on one Intel Xeon Platinum 8273CL core (2.2 GHz), with double precision and 2 TB of available memory. The measurements exclude index construction. The C++17 implementations use GCC 11.4 with `-O3 -march=native` and share finite-field arithmetic and subspace indexing. Input weights are uniform in $[-1, 1]$ with fixed pseudorandom seeds.

For the zeta transform Z_n, we ran each algorithm three times and kept its shortest measured running time. We repeated this procedure 3–21 times, changing which algorithm ran first each time. We report the median of the retained measured running times and compute each speedup from the two times in the same repetition. The figure shows the median and quartiles of these speedups. For the incidence transform and Grassmann kernels, our measured running times are medians of 3–15 and 3–25 runs, respectively, after a warm-up. Long incidence comparisons were run once. For repeated Poisson runs, the measured running time is the sum of the medians for its three stages. Operation counts use real arithmetic, including the operations needed for complex Fourier arithmetic for odd q. Memory figures count the specified tables, workspace, and indices in decimal GB or TB, not peak process memory.

			ours	operation count / ours			time / ours	
q	(n,a,b)	S	time	direct	covering	FFT bound	covering	Poisson
2	$(16,1,15)$	$1.3\cdot10^5$	1.9 ms	1600	$3.0\cdot10^{16}$	0.76	–	0.68
2	$(14,2,12)$	$8.9\cdot10^7$	2.1 s	$7.1\cdot10^4$	$3.5\cdot10^8$	4.3	–	4.5
3	$(8,2,6)^\dagger$	$1.8\cdot10^6$	32 ms	120	82	8.4	550	42
4	$(8,2,6)^\dagger$	$4.8\cdot10^7$	1.0 s	2700	1400	170	–	130
4	$(9,2,7)$	$7.6\cdot10^8$	20 s	$3.9\cdot10^4$	$5.6\cdot10^4$	170	–	190
8	$(6,2,4)$	$3.9\cdot10^7$	810 ms	140	34	3700	140	5300
8	$(7,2,5)$	$2.5\cdot10^9$	80 s	7500	1100	3600	–	–
2	$(10,3,7)$	$1.3\cdot10^7$	260 ms	300	33	130	84	140
2	$(11,3,8)$	$1.0\cdot10^8$	2.9 s	2200	210	130	–	130
2	$(12,3,9)^\dagger$	$8.2\cdot10^8$	28 s	$1.6\cdot10^4$	1900	130	–	160

Table 2: Incidence transforms $\mathsf{I}_n^{a,b}$ with $b=n-a$ and $S=|\operatorname{Gr}_q(n,a)|+|\operatorname{Gr}_q(n,b)|$. Ratios above one favor our algorithm. Operation counts are divided by our count of real arithmetic operations. "FFT bound" is the conservative count for Poisson summation described in the text. A dash means the comparison was not run because of memory or indexing limits: the covering-pair runs allow 400 GB of estimated memory and 32-bit table indices. † marks the family $(4r,r,3r)$ of theorem 7.

Zeta transform. Our zeta transform uses the Fourier recursion without a covering-pair base case. On $\mathbb{F}_2^{10}$, the covering-pair algorithm uses 0.97 times as many arithmetic operations as ours, but its measured running time is 2.0 times ours. On $\mathbb{F}_{32}^5$, with $N_{32}(5)\approx 2.2\cdot10^9$, our measured running time is 145 s, whereas the covering-pair algorithm would need 5 TB just to store one index per covering pair. To estimate its running time, we processed about 2% of the covering pairs from 64 evenly spaced ranges of output subspaces, using the full value table, and scaled by the ratio of total to sampled pairs. This estimate is 95 times our measured running time, and the figure marks it separately from measured running times. Applying the same procedure on $\mathbb{F}_{16}^5$ and $\mathbb{F}_8^6$ underestimates their measured running times by 11% and 16%, respectively, but does not determine the error on $\mathbb{F}_{32}^5$. The operation count ratios in the figure are computed from the recursion and exact covering-pair counts, including instances too large to run.

Incidence between separated dimensions. For $\mathsf{I}_n^{a,b}$, direct summation processes $|\operatorname{Gr}_q(n,b)|\,|\operatorname{Gr}_q(b,a)|$ incident pairs, while the covering-pair algorithm passes through the intermediate dimensions, using $\sum_{j=a+1}^{b}|\operatorname{Gr}_q(n,j)|[j]_q$ additions. Table 2 compares these operation counts with ours.

For Poisson summation, write $t=a=n-b$, as in all instances in the table. Place $f(U)$ at the ordered reduced row-echelon basis of U in a table F on $(\mathbb{F}_q^n)^t$, and set its other entries to zero. Every $U\le W$ has exactly one such basis in W^t, so Poisson summation [MP13, Theorem 10.1.16] gives

$$(\mathsf{I}_n^{t,n-t}f)(W)=\sum_{x\in W^t}F(x)=q^{-t^2}\sum_{y\in(W^\circ)^t}\widehat{F}(y).$$

The Fourier table has $q^{tn}=\Theta_q(q^{t^2}S)$ entries, by (24). For $q=p^m$, the table's "FFT bound" is the conservative count $tmnq^{tn}$, charging one operation per entry per base-p coordinate. It omits the final sums and undercounts the real arithmetic for odd p. The measured running time includes constructing F, taking its Fourier transform, and summing the required frequencies.

For $\mathsf{I}_7^{2,5}$ over $\mathbb{F}_8$, our tables, workspace, and indices use about 70 GB. The intermediate layers alone require 1.3 TB of values before storing any covering pairs, while the Fourier table for Poisson summation requires 35 TB, so neither comparison algorithm fits in memory. When $t=1$, the Poisson

Graph	$N = \lvert \mathrm{Gr}_q(n,k) \rvert$	ours time	time / ours: distance-matrix recurrence	time / ours: two covering-pair transforms
$J_2(10,5)$	$1.1 \cdot 10^8$	17 s	6.2	1.23
$J_3(8,4)$	$7.6 \cdot 10^7$	10 s	8.0	–
$J_5(6,3)$	$2.6 \cdot 10^6$	170 ms	4.8	–
$J_8(6,3)$	$1.6 \cdot 10^8$	17 s	22	6.6

Table 3: Large Grassmann kernel instances, with $a_i = 2^{-i}$. The last column replaces the fast transforms in (17) by covering-pair transforms and is reported only in characteristic two. Measured running times exclude index construction.

table has only q^n entries and each final sum has q terms, explaining its shorter measured running time in the first row.

Grassmann kernels. For $L \in \mathrm{Gr}_q(n,k-1)$ and $U \in \mathrm{Gr}_q(n,k)$, let $B(L,U) = \mathbf{1}[L \le U]$. The identity [MOR03, Section 5.3]

$$D_1 = B^{\mathsf{T}} B - [k]_q I$$

reduces each adjacency multiplication to two incidence sums, with $2[k]_q \lvert \mathrm{Gr}_q(n,k) \rvert$ additions and a diagonal correction, without storing the adjacency matrix. The recurrence uses real arithmetic even for odd q, and we conservatively count only its $2d[k]_q N$ incidence additions, omitting the scalar work in the recurrence and weighted sum. Our count includes both transforms, the intermediate scalings, and the coefficient calculation. A complex scaling counts as two real multiplications.

Checking the outputs. We check direct sums on all outputs of small instances and sampled outputs of large ones (200 per incidence instance), and compare all outputs whenever both methods produced all outputs. For computed values y, reference values y_{ref}, and each set T of checked outputs, the relative discrepancy satisfies

$$\frac{\max_{W \in T} \lvert y(W) - y_{\mathrm{ref}}(W) \rvert}{\max_{W \in T} \lvert y_{\mathrm{ref}}(W) \rvert} < 10^{-13}.$$

These checks do not give a numerical-stability guarantee. Exact arithmetic is covered by sections 3.4 and D.

F A matching lower bound for incidence

Theorem 7. *For $r \ge 1$ and $v = \lvert \mathrm{Gr}_q(4r,r) \rvert$, applying $\mathsf{I}_{4r}^{r,3r}$ over $\mathbb{C}$ requires $\Omega_q(v \log v)$ gates of the form $z = \alpha x + \beta y$, $\lvert\alpha\rvert, \lvert\beta\rvert \le 1$. The algorithm of theorem 2 attains this bound up to a constant factor depending on q.*

Proof. We lower-bound the determinant and apply Morgenstern's bounded-coefficient argument [Mor73]. Put $I = \mathsf{I}_{4r}^{r,3r}$. For r-spaces U, W with $t = \dim(U \cap W)$, quotienting by $U + W$ gives

$$(I^{\mathsf{T}} I)(U,W) = \lvert \mathrm{Gr}_q(2r+t, r+t) \rvert = \lvert \mathrm{Gr}_q(2r+t, r) \rvert.$$

Count the r-spaces in $\mathbb{F}_q^{2r} \oplus (U \cap W)$ by their projection $L \le U \cap W$. For a fixed j-dimensional L, choose an $(r-j)$-space $A \le \mathbb{F}_q^{2r}$ and a map $L \to \mathbb{F}_q^{2r}/A$. There are $c_j = \lvert \mathrm{Gr}_q(2r, r-j) \rvert q^{j(r+j)}$

choices. Thus $(I^{\mathsf{T}}I)(U,W) = \sum_{L \le U \cap W} c_{\dim L}$, and every real vector $x = (x_U)_{U \in \mathrm{Gr}_q(4r,r)}$ satisfies

$$\|Ix\|_2^2 = \sum_{\substack{L \le \mathbb{F}_q^{4r} \\ \dim L \le r}} c_{\dim L} \left(\sum_{\substack{U \in \mathrm{Gr}_q(4r,r) \\ L \le U}} x_U \right)^2 \ge c_r \|x\|_2^2 = q^{2r^2} \|x\|_2^2.$$

The inequality keeps just the terms with $\dim L = r$. Every singular value of the square matrix I is therefore at least q^{r^2}. Since $\log v = \Theta_q(r^2)$ by (24), $\log |\det I| \ge vr^2 \log q = \Omega_q(v \log v)$.

Regard each wire as a row of coefficients in the v inputs. Initially every v-row determinant has absolute value at most one. A gate $\alpha x + \beta y$ with $|\alpha|, |\beta| \le 1$ increases the largest such determinant by at most a factor of two, by multilinearity. Thus at least $\log_2 |\det I|$ gates are needed. The upper algorithm uses only additions, roots of unity, and Fourier normalizations of modulus at most one, so it fits this model. This is a lower bound for the stated incidence matrices, not for unrestricted constants or for the full zeta matrix, whose determinant is one. □

G The counter reduction

We adapt the monotone-counter reduction of [FGL+19], replacing binary variable identifiers by subspaces of $\mathbb{F}_2^{2d}$. A constraint uses colors of a specified type, and a single clique will test all these colors together.

Theorem 8. *Assuming* ETH*, for every computable function f and fixed $\alpha \ge 0$, there is no algorithm deciding whether $\chi(G) \le \eta$ in time*

$$f(r) N^{\alpha + 2^{r^2/4 - 17r}}$$

on N-vertex graphs supplied with a linear layout of width r, where η is part of the input. The same exclusion holds with a supplied rank-decomposition of width r.

G.1 Monotone counters and the register

An instance of 4-Monotone min-CSP has k variables with values in $\{0, \dots, n\}$, a budget W, and constraints $\mathcal{C}$. A constraint C involves a set X_C of at most four variables and a nonempty family $\mathcal{R}_C$ of *mini-constraints*. Each $R \in \mathcal{R}_C$ requires $\alpha(x) \ge c_{R,x}$ for every $x \in X_C$, where $c_{R,x} \in \{0, \dots, n\}$. We seek an assignment of cost $\sum_x \alpha(x) \le W$ satisfying at least one mini-constraint of each C. Increasing values preserves satisfaction.

Lemma 4 ([FGL+19], Lemma 3.1 and its construction)**.** *Under* ETH*, for any computable f, this problem has no $f(k)n^{o(k)}$ algorithm, even when $|\mathcal{C}| = O(k^2 n^2)$ and $L := \max(1, \max_C |\mathcal{R}_C|) \le \max(n, 4)$.*

The construction in [FGL+19, Lemma 3.1] uses $2K$ counters for a K-part Multicolored Clique instance with n vertices per part. It has K constraints with n mini-constraints and at most $\binom{K}{2} n^2$ constraints with at most four mini-constraints. Discard mini-constraints with thresholds greater than n and pad missing inequalities with threshold zero. Empty families give an immediate rejection, while empty scopes are tautologies and can be deleted. We may handle $k < 2$, $n = 0$, and budgets outside $[0, kn]$ directly.

Put $d = \lceil \sqrt{\log_2 k} \rceil$, $M = 2^{d^2}$, $h = 2^d$, and $Q = Mn$. Pad with unused variables and index the resulting M variables by matrices $A \in \mathbb{F}_2^{d \times d}$. Make d disjoint Q-cliques B_i, with Q/h vertices of

each label (e_i, a), $a \in \mathbb{F}_2^d$, in B_i. Also make a disjoint clique B_* of size $Q - W$. The construction below forces all rows to use a common Q-color palette containing the colors of B_*. Each color occurs once in each row, and its labels $(e_i, A^\mathsf{T} e_i)$ define its matrix type A. The W colors absent from B_* are called *active*, and their number of type A represents x_A.

To identify a color's type, associate A with the subspace $S_A = \{(x, A^\mathsf{T} x) : x \in \mathbb{F}_2^d\}$ and the basis $t_{A,j} = (Ae_j, e_j)$, $1 \le j \le d$, of $S_A^\perp$. A color of type T passes every orthogonality test exactly when $T = A$, since

$$(e_i, T^\mathsf{T} e_i)^\mathsf{T} t_{A,j} = T_{ij} + A_{ij}. \tag{32}$$

Every family of integers $0 \le \alpha_A \le n$ with $\sum_A \alpha_A = W$ can be realized as the active color counts. Allocate α_A distinct colors to type A and place them on its label in every row. Exactly M/h matrices have a prescribed row, so no label receives more than $(M/h)n = Q/h$ colors. In each row independently, fill its remaining $Q - W$ vertices bijectively with the same $Q - W$ new colors, and use those on B_*. There are no edges between rows. A satisfying assignment of cost below W can be increased to cost W: the total capacity is $Mn \ge W$, and increasing values preserves every source constraint.

G.2 Constraint gadgets

For each threshold $x_A \ge c$, an n-clique will need at least c active colors of type A. A selector chooses a mini-constraint and makes at most $n - c$ other colors available to this clique. For each C and $x \in X_C$, make an n-clique $A_{C,x}$. For each $R \in \mathcal{R}_C$ and $x \in X_C$, make an independent set $I_{C,R,x}$ of size $n - c_{R,x}$. Write $I_{C,R} = \bigcup_x I_{C,R,x}$, $I_{C,x} = \bigcup_R I_{C,R,x}$, and $I_C = \bigcup_R I_{C,R}$. Join distinct $I_{C,R}$ completely for fixed C, and join every $I_{C,R,x}$ completely to $A_{C,x}$. There are no other edges among these A- and I-vertices.

Make a selector s_C for each C and a private anchor m_b for each I-vertex b. All selectors and anchors form a clique D, completely joined to every B_i and to B_*. Join s_C to every A- and I-vertex outside I_C. If $b \in I_{C,R,x}$, join m_b to every A- and I-vertex outside $A_{C,x} \cup \{b\}$. Denote the anchors belonging to $I_{C,x}$ by $\mathcal{M}_{C,x} = \{m_b : b \in I_{C,x}\}$. Join all A- and I-vertices to B_*, and all I-vertices to every B_i. Set the target number of colors to

$$\eta = Q + |D|. \tag{33}$$

In an η-coloring, the η-cliques $B_i \cup D$ force each B_i to use exactly the complement of the D-colors. Since B_* is complete to D, its colors lie in this common register palette.

It remains to restrict the register colors on A_{C,x_A} to type A. The test needs to copy only the set of colors on this clique. Call the vertices constructed so far *old vertices*. The following cliques have only the edges specified below. For each A_{C,x_A}, make d n-cliques $A^{(1)} = A_{C,x_A}, A^{(2)}, \ldots, A^{(d)}$, of which only the first keeps the edges already specified. Join every vertex of $A^{(j)}$ to a register vertex of label v exactly when $v^\mathsf{T} t_{A,j} = 1$. Add a private $(\eta - n)$-clique E_{C,x_A}, complete to these d cliques and with no other neighbors. These test cliques together with E_{C,x_A} form a *palette block*.

The private clique leaves exactly n colors, so every $A^{(j)}$ uses the same palette. Each register color in this palette passes all the tests in (32), and therefore has type A. Conversely, any coloring satisfying the old edges and using only type-A register colors on $A^{(1)}$ extends: use its palette on every test clique and all other colors on E_{C,x_A}. Colors from D have no register conflicts. After fixing the register and D, identify anchors with their colors and let P_A be the active colors of type A. The remaining problem is exactly list coloring of the old A/I graph with

$$L(b) = \{s_C, m_b\} \quad (b \in I_C), \qquad L(a) = P_A \cup \mathcal{M}_{C,x_A} \quad (a \in A_{C,x_A}).$$

The specified edges give precisely these lists, and every such coloring extends independently to the palette blocks.

G.3 Correctness

Lemma 5. *The graph is η-colorable if and only if the source instance is satisfiable.*

Proof. Given an η-coloring, let z_A count the active colors of type A, so $\sum_A z_A = W$. Since distinct $I_{C,R}$ are completely joined, at most one mini-constraint R has a vertex using the selector color. Choose that R, or any R if none does. Every vertex belonging to another mini-constraint uses its private anchor color. Its complete join to $A_{C,x}$ blocks that color on the whole clique. For $x_A \in X_C$, at most $|I_{C,R,x_A}| = n - c_{R,x_A}$ anchor colors therefore remain available to A_{C,x_A}. Its n vertices have distinct colors, so

$$n \le (n - c_{R,x_A}) + z_A, \qquad z_A \ge c_{R,x_A}.$$

Consequently $\alpha(x_A) = \min(n, z_A)$ satisfies every constraint and has total cost at most W. Restrict it to the original variables.

Conversely, increase a satisfying padded assignment to cost W and realize it in the register, using colors disjoint from those of D. Choose a satisfied mini-constraint R for each C. Color $I_{C,R}$ with the selector color and every other b with its private color. Color A_{C,x_A} with the $n - c_{R,x_A}$ private colors from I_{C,R,x_A} and c_{R,x_A} distinct active colors of type A. This is a proper list coloring: different I-groups use disjoint colors, and A uses no unchosen group's private color. Extend the palette blocks as above. Empty I-sets cause no change. □

G.4 The linear layout

Lemma 6. *The construction supplies a linear layout of cut-rank at most $2d + 12$.*

Proof. Order the row cliques first, then B_*. Process constraints consecutively. For each C, put s_C first, then its palette blocks one at a time, then its mini-constraints one at a time. Within a palette block, order the whole cliques $A^{(1)}, \dots, A^{(d)}, E_{C,x}$ consecutively. Within a mini-constraint, put each pair m_b, b consecutively.

The register-to-test edges have rank at most $2d$: they factor through the register labels in $\mathbb{F}_2^{2d}$, and all test vertices follow the register. The universal columns to D and I lie in the same space, since the sum of a register label's first d coordinates is one. During the register prefix at most one row clique is split, and during B_* its processed vertices have one common future neighborhood. These initial cuts therefore have rank at most $2d + 1$.

Consider a later cut, in the part for constraint C. Remove the register edges, the edges within test cliques, and the private-clique edges. The remaining nonzero rows have at most ten different future neighborhoods:

- D-vertices whose exceptions are in the past, and B_*, see all future old vertices, whereas A/I-vertices of completed constraints see exactly future D-vertices.
- The current selector gives one further neighborhood.
- Current A-vertices give at most four, one for each $x \in X_C$: they see future D outside $\mathcal{M}_{C,x}$ and future $I_{C,x}$.
- Current I-vertices give at most two, according to whether their mini-constraint is current or already finished.
- A split pair m_b, b adds at most one exceptional anchor.

Indeed, all A-cliques precede the anchors, and each finished anchor has both its A-clique and its exceptional vertex b in the past. Thus the remaining rank is at most $2+1+4+2+1=10$. At most one test clique is split, contributing rank one. At most one palette block is split, and its crossing private edges form a complete bipartite graph, contributing rank one. Adding back the register edges gives cut-rank at most $2d+10+1+1$. □

G.5 Size and the ETH exponent

Proof of theorem 8. Write $a=\sum_C |X_C|$ and $p=|\mathcal{C}|$. There are a palette blocks, each with $dn+\eta-n$ vertices, and $|D|=O(p+anL)$ anchors, so

$$N=O\big(dMn+p+anL+a(\eta+dn)\big)\le g(k)n^6$$

for a computable g: use $p=O(k^2n^2)$, $a\le 4p$, and $L\le 4n$ from lemma 4. Constructing all edges and the layout takes $g_1(k)n^{O(1)}$ time with an absolute polynomial degree. Put $s=\sqrt{\log_2 k}$. The actual layout width satisfies $0\le r\le 2\lceil s\rceil+12\le 2s+14$. For the convex function $b(t)=t^2/4-17t$,

$$2^{b(r)}\le \max\{1,2^{b(2s+14)}\}=\max\{1,k2^{-20s-189}\}=o(k).$$

Bound $f(r)$ by the maximum of $1, f(j)$ over the integers $0\le j\le 2\lceil s\rceil+12$. Combining the hypothetical running time with the construction cost and absorbing all factors depending only on k gives $F(k)n^{o(k)}$, contrary to lemma 4. Finally, the graph contains a Q-clique with $Q\ge 2$, so $r\ge 1$. The layout's caterpillar rank-decomposition has the same nonleaf cuts and singleton cuts of rank at most one, and therefore the same width. □

H Fixed-palette decision and counting lower bounds

Theorem 9. *Assuming* ETH*, for every fixed $c\ge 3$, deciding c-colorability has no $2^{o(r^2)}N^{O(1)}$-time algorithm on N-vertex graphs supplied with a linear layout of width at most r. The same exclusion holds for supplied rank-decompositions.*

Theorem 10. *Assume* #ETH*. For every fixed $c\ge 3$, exact counting of proper c-colorings has no deterministic $2^{o(r^2)}N^{O(1)}$ algorithm, even on graphs supplied with a linear layout of width at most r.*

One reduction proves both statements: each satisfying assignment will give the same known number of labeled three-colorings. Here #ETH excludes $2^{o(n)}\,\mathrm{poly}(m)$ time for counting satisfying assignments of 3-CNF formulas with n variables and m clauses [DHM+14]. Let φ have k^2 variables in k rows of length k, with $k\ge 1$, and m clauses. Repeat literals in shorter nonempty clauses so each clause has three literal occurrences. Write J_i for the set of rows containing variables of clause i.

Making the row assignments agree. Create a triangle with vertices and colors named T, F_A, F_B. An input is *selected* when it has color T. For every row j, create a base group of inputs $a_{0,j,u}$, one for each assignment $u\in\mathbb{F}_2^k$ to that row. For each clause i and $j\in J_i$, create another group $a_{i,j,u}$ with the same possible assignments. All these inputs have label $(e_j,u)\in\mathbb{F}_2^{2k}$ and are adjacent to F_B, restricting their colors to $\{T,F_A\}$.

To make different copies agree, introduce one group of inputs $b_{h,v}$, $v\in\mathbb{F}_2^k$, for each column $h\in[k]$. Give these inputs labels (v,e_h) and join them to F_A, restricting their colors to $\{T,F_B\}$. Join an A input to a B input when their labels have dot product one. Since their color sets intersect only in T, selected labels must satisfy

$$(e_j,u)^{\mathsf{T}}(v,e_h)=0 \quad\Longleftrightarrow\quad u_h=v_j.$$

Suppose every group selects at least one input. A selected vector v_h in each column group fixes all coordinates of every selected row assignment: $(u_j)_h = (v_h)_j$. Thus each row group selects exactly one assignment, identically in all its copies. The base row groups in turn fix every selected column assignment, so each column group also selects exactly one input. Conversely, any matrix of row assignments u_j gives exactly these selections, with the v_h as its columns. Base groups ensure that variables absent from all clauses are still represented.

Enforcing a nonempty choice. The preceding argument already excludes multiple selections once each group selects at least one input. An odd cycle suffices to enforce this remaining condition. For a group $x_1, \dots, x_t$, where $t = 2^k$, let F be its false color, either F_A or F_B. Add a fresh cycle $w_0 w_1 \cdots w_t w_0$, join w_0 to F, and join w_i to x_i for $1 \le i \le t$.

If every input is false, all cycle vertices avoid F, which would two-color an odd cycle. If exactly x_j is selected, every cycle vertex except w_j still avoids F. Hence w_j must have color F, and the remaining path alternates the other two colors in exactly two ways. Thus every coloring selects at least one input in each group, and every consistent singleton selection has exactly two extensions on its cycle. No claim about larger selections is needed.

Testing a clause with a fixed number of extensions. For each literal occurrence, make an adjacent pair $c, \bar{c}$, both adjacent to F_A, so they have opposite colors in $\{T, F_B\}$. Join c to the clause-local row inputs that falsify the literal, and $\bar{c}$ to those that satisfy it. The unique selected input then forces $c = T$ precisely when the literal is true, while unselected inputs have color F_A and impose no further restriction.

For three terminals in $\{T, F_B\}$, a fresh triangle with one vertex adjacent to each terminal enforces the not-all-equal relation NAE. Equal terminal colors leave only two colors for the triangle. For mixed colors, the vertex opposite the singleton terminal color must use the repeated color, and the other two vertices can use the remaining colors in either order. Thus every valid input has exactly two extensions, also for repeated terminals or constants.

For a clause's three literal vertices a, b, c, introduce an adjacent pair $s, \bar{s}$, both adjacent to F_A, and four private NAE triangles imposing

$$\begin{aligned} &\mathrm{NAE}(\bar{a}, s, F_B), \qquad \mathrm{NAE}(\bar{b}, s, F_B), \\ &\mathrm{NAE}(a, b, \bar{s}), \qquad \mathrm{NAE}(s, c, F_B). \end{aligned} \tag{34}$$

The first two conditions say that a or b being true forces s true. The third forces s false when both are false, and the last requires $s \vee c$. Thus $s = a \vee b$ is uniquely determined, and the clause has exactly $2^4 = 16$ extensions when satisfied and none otherwise. All auxiliary vertices of the cycles and triangles are private, and there are no edges other than those specified.

Every proper coloring now determines a satisfying assignment. Conversely, a satisfying assignment fixes all inputs, literal pairs, and auxiliary pairs once the palette colors are fixed. There are $g = 2k + \sum_i |J_i|$ groups, each with two cycle extensions, and each clause contributes sixteen extensions. Therefore

$$\#\text{3-Color}(G_\varphi) = 6 \cdot 2^g \cdot 16^m \, \#\mathrm{SAT}(\varphi), \tag{35}$$

where six counts the labeled palette colorings.

The supplied layout and its size. Order the palette first, then the B groups, then the base A groups, then each clause's local A groups followed by its clause test. Within a group, use the order $w_0, x_1, w_1, \dots, x_t, w_t$. Within a clause test, put the four literal and auxiliary pairs first and then the four triangles, one at a time.

For any prefix cut, the AB block has rank at most $2k$ by its label factorization: all B inputs precede all A inputs. Palette edges have rank at most three. Edges from row inputs to literal pairs have at most six future columns, all in one clause. For the remaining local edges, only one group or clause test can straddle the cut. In a group, at most w_0, the last cycle vertex, and the current input have later local neighbors. In a clause test, at most eight vertices in the literal and auxiliary pairs and two vertices of the unfinished triangle have later neighbors. Subadditivity therefore bounds every cut-rank by $2k + 3 + 6 + 10 = 2k + 19$, including cuts inside pairs and triangles.

The graph has $3 + g(2^{k+1} + 1) + 20m$ vertices: a group contributes 2^k inputs and $2^k + 1$ cycle vertices, and a clause contributes eight vertices in the literal and auxiliary pairs and twelve triangle vertices. Its $O((k + m)k2^{2k})$ edges and the layout are constructible in $2^{O(k)}\,\mathrm{poly}(m)$ time. For an n-variable formula, take $k = \lceil\sqrt{n}\rceil$ and pad to $k^2 = n + O(\sqrt{n})$ variables. For counting, divide out both the factor in (35) and the 2^{k^2-n} assignments to the added variables. Empty clauses, empty formulas, and bounded cases are handled directly. A $2^{o(r^2)}N^{O(1)}$ algorithm would therefore decide or count SAT in $2^{o(n)}\,\mathrm{poly}(m)$ time, contrary to ETH or #ETH [IP01, DHM+14]. The count and divisors have $O(|V(G_\varphi)|)$ bits, so exact division fits this bound.

For fixed $c > 3$, add a universal $(c-3)$-clique and place it first. It consumes exactly $c-3$ colors, adds at most $c-3$ to the layout width, and multiplies the labeled count by $c!/3!$. Finally, a linear layout gives a caterpillar rank-decomposition with the same nonleaf cuts and leaf cuts of rank at most one. This proves both theorems.

AI methodology statement. ChatGPT and Claude assisted with literature checks, proofs, editing, and verification code for the transforms, linear algebra, and coloring reductions. Claude Code also assisted with the Lean 4 formalization described in the introduction. The authors are responsible for the claims and attribution. These checks are not external expert review.

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